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On the Theory and Design of Acoustic Resonators

Uno Ingard



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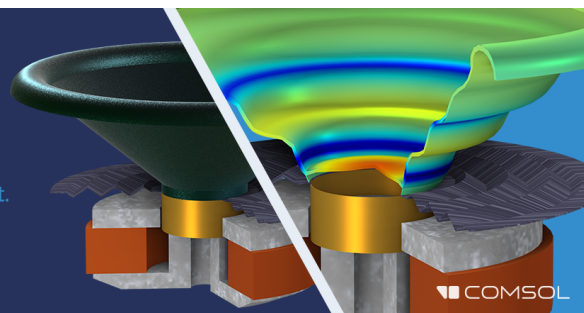
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On the Theory and Design of Acoustic Resonators*†

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Absorption and scattering from resonators in a free field as well as in walls are discussed. The effect of different aperture geometries on the resonance frequency of resonators is considered and illustrated by examples. Considering losses due to viscosity, heat conduction, and radiation, the optimum design for maximum resonance absorption is analyzed, and the results are expressed in terms of design charts. Nonlinear effects on the absorption and resonance frequency are also included, and a discussion of the onset of turbulence is presented.

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I. INTRODUCTION

THE studies of acoustic cavity resonators initiated by Helmholtz and Rayleigh¹ have been followed by numerous investigations on the subject.² It is still a challenging research topic not only because there remain "untouched" aspects of the problem but also because of the discrepancies between theory and experiments that have been reported repeatedly in recent years.³ These discrepancies have involved not only the

¹Rayleigh, *Theory of Sound* (MacMillan and Company, Ltd., London, 1940), Vol. II, p. 170.

²P. O. Pedersen, *Lydtekniske Undersøgelser Ingeniør-Videusksplige Skrifter No. 5*, Copenhagen (1940). Kosten and Zwikker, *Sound Absorbing Materials* (Elsevier Publishing Company, Inc., New York, 1949). This book contains further references on the subject.

³A summary of references on this aspect can be found in Bolt, Labate, Ingard, *J. Acoust. Soc. Am.* **21**, 94 (1949).

* The major portion of this work was done during 1949–1950 when the author held a fellowship in acoustics sponsored by the Acoustical Materials Association.

† Presented at the Philadelphia meeting of the Acoustical Society of America, May 7–9, 1953.

resonance frequency but in particular the absorption and by the resonator or its damping.

The major purpose of the present paper is to discuss some properties of a resonator in a free field or in a wall in regard to scattering and absorption of an incident plane wave. When the dimensions of the resonator in a free field are very small compared to the wavelength of the incident wave, the expressions for the scattered and absorbed power are exactly analogous to those for the electric dipole when written in terms of internal and radiation impedance.⁴ The resonator can then be represented by a series resonant electric circuit in which the resistance is the sum of the internal and the radiation resistance. The driving voltage on the circuit corresponds to the "blocked" pressure at the resonator which equals the pressure P_0 of the incident wave for a resonator in a free field and $2P_0$ for a resonator in a wall. In most practical cases this long-wavelength approximation is often satisfactory.

II. SCATTERING AND ABSORPTION CROSS SECTIONS

For sufficiently low frequencies, the powers scattered and absorbed by a resonator in a free field are

$$W_s = P_0^2 \theta_r A_0 / 2\rho c |\zeta|^2 \quad \text{and} \quad W_a = P_0^2 \theta_i A_0 / 2\rho c |\zeta|^2,$$

respectively. The corresponding scattering and absorption cross sections σ_s and σ_a are

$$\sigma_s = \frac{\theta_r}{|\zeta|^2} A_0 \quad (1)$$

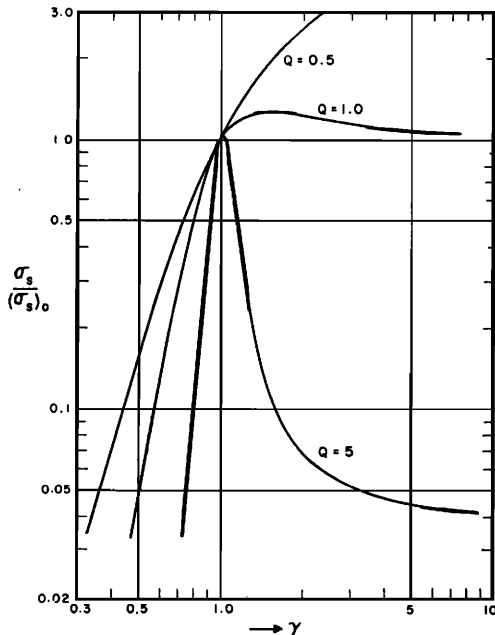


FIG. 1. The frequency dependence of the scattering cross section for a resonator in a free field for different values of Q .

⁴ P. O. Pedersen, reference 2.

$$\sigma_a = \frac{\theta_i}{|\zeta|^2} A_0, \quad (2)$$

where θ_r = the normalized specific radiation resistance of the resonator aperture, θ_i = the corresponding aperture resistance due to internal friction, A_0 = the area of the aperture, and ζ = the total normalized specific impedance of the aperture. At resonance the absorption cross section σ_a reduces to

$$\sigma_a = \frac{\theta_i}{(\theta_i + \theta_r)^2} A_0 = \frac{\lambda_0^2}{4\pi} \frac{4\theta_i/\theta_r}{(1 + \theta_i/\theta_r)^2}, \quad (3)$$

in which has been introduced the radiation resistance at low frequencies $\theta_r = A_0 \pi / \lambda_0^2$ for a piston in a sphere, λ_0 being the resonance wavelength. A similar expression holds for the scattering cross section. The absorption cross section has a maximum equal to $\lambda^2 / 4\pi = 0.0796 \lambda^2$, when $\theta_i = \theta_r$, which is the same as for the corresponding electric dipole scatterer.⁵ Under these optimum conditions the scattering cross section has the same value as the absorption cross section,

$$(\sigma_a)_{\max} = (\sigma_s)_{\max} \simeq \lambda_0^2 / 4\pi.$$

In the present paper the major effort is focused on the problem of the design of a resonator in a free field and in a wall for maximum resonance absorption. It is thus shown that for a given resonance frequency there exists an optimum volume of the resonator at which the resonance absorption cross section (and the Q value) is a maximum.

The analysis shows that a resonator with smooth hard surfaces and without any additional resistance in the resonator neck in the form of screens, etc. often has a Q value of the order of 30 to 40. Such a Q value is generally too high for the practical use of the resonator as an absorption unit for sound extending over a large frequency range. In such a case additional resistance must be introduced, the proper amount depending on the absorption characteristics desired. The introduction of the resistance means a reduction of the resonance absorption cross section unless the optimum design is restored by increasing the volume of the resonator (assuming frequency kept constant). Usually a compromise design has to be made which results in a resonator whose resonance absorption in a free field may be considerably smaller than $\lambda^2 / 4\pi$.

The Frequency Dependence of Absorption and Scattering Cross Sections

So far we have been concerned only with the behavior of the resonator at resonance. The frequency dependence is obtained from the original equations (1) and (2). Although the total resistance $\theta = \theta_i + \theta_r$ is frequency

⁵ J. C. Slater, *Microwave Transmission* (McGraw-Hill Book Company, Inc., New York, 1942), p. 245.

dependent, it is a good approximation to set

$$\zeta = (\theta_i + \theta_r)_0 [1 - iQ(\gamma - 1/\gamma)],$$

where $(\theta_i + \theta_r)_0$ is the value at the resonance frequency ω_0 of the resonator and $Q = k_0(t + \delta)/(\theta_i + \theta_r)_0$. The absorption and scattering cross sections can then be written

$$\frac{\sigma_a}{(\sigma_a)_0} = \frac{1}{1 + Q^2(\gamma - 1/\gamma)^2},$$

$$\frac{\sigma_s}{(\sigma_s)_0} = \frac{\gamma^2}{1 + Q^2(\gamma - 1/\gamma)^2},$$

where $(\sigma_a)_0$ and $(\sigma_s)_0$ are the values of absorption and scattering cross section at the resonance frequency corresponding to $\gamma = \omega/\omega_0 = 1$. As will be shown in Sec. V, $\theta_r/\theta_i = (a/a_1)^6$ and the relation between $(\sigma_a)_0$ and $(\sigma_s)_0$ is

$$(\sigma_s)_0 = \left(\frac{a}{a_1}\right)^6 (\sigma_a)_0 = \left(\frac{V}{V_1}\right)^2 (\sigma_a)_0,$$

where a/a_1 is the ratio between the resonator radius a and the radius a_1 for optimum design and V/V_1 is the corresponding ratio between the volumes.

It is interesting to notice that the scattering cross section reaches its maximum value at a frequency above resonance. In fact, the maximum value occurs when

$$\gamma = \left(1 - \frac{1}{2Q^2}\right)^{-1/2},$$

and the corresponding maximum scattering cross section is

$$(\sigma_s)_{\max} = (\sigma_s)_0 \left(1 - \frac{1}{4Q^2}\right)^{-1}.$$

A maximum of σ_s is obtained only for $Q > 1/\sqrt{2}$; when $Q < 1/\sqrt{2}$ the scattering increases monotonically with increasing γ , as can be seen in Fig. 1, where the scattering cross section is plotted as a function of γ for different values of Q . Correspondingly the absorption cross section is shown in Fig. 2.

The analysis thus presented applies only to the fundamental resonance, and since it is evident that the higher resonances will also cause absorption and scattering, the frequency response in Figs. 1 and 2 must be modified accordingly.

III. THE RESONANCE FREQUENCY

The expression for the resonance frequency ν_0 of a resonator is commonly written in the form

$$\nu_0 = \frac{c}{2\pi} \left[\frac{A}{V(t + \delta)} \right]^{1/2},$$

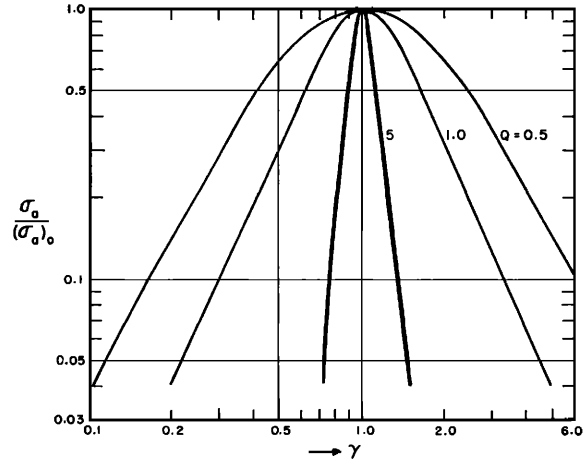


FIG. 2. The frequency dependence of the absorption cross section for a resonator in a free field for different values of Q .

where A = the area of the aperture, V = the volume of the cavity, t = the length of the aperture neck, and δ = the "end correction" of the aperture neck. Usually the end correction is indiscriminately taken as the mass end correction for a plane circular piston in an infinite plane, which equals $(16/3\pi) \cdot r_0 \approx 1.7r_0$,⁶ where r_0 is the radius of the piston or the circular aperture. To make it applicable for an arbitrary aperture, the end correction is sometimes written $\delta = 0.96(A)^{1/2}$.

Although this end correction represents a good approximation when the aperture is very small compared to the cross dimension of the resonator cavity, it can be in considerable error under less extreme conditions. A careful analysis should actually consider different end corrections on the two sides of the aperture so that $\delta = \delta_e + \delta_i$, i.e., the sum of an exterior end correction δ_e and an interior correction δ_i .

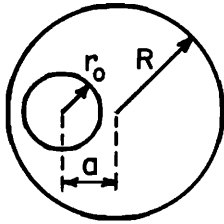
The rigorous calculation of δ_i and δ_e for a given resonator geometry is quite involved and can in general hardly be done. To find these quantities the particular diffraction problem dealing with the transmission of sound through the resonator aperture must be solved. The difficulty arises from the fact that the velocity distribution in the aperture is not known *a priori*. It is known, however, that the end correction is stationary (minimum) with respect to variation of the velocity distribution about the correct one. This feature has been used to determine end corrections somewhat more accurately than obtained simply on the assumption of a uniform distribution.⁷ On the other hand, it should be borne in mind that an analysis of that kind neglects the effect of viscosity which in many cases may modify the idealized velocity distribution considerably. The linear wave analysis of the problem neglecting velocity leads to an infinite velocity at sharp corners. Actually, this infinity is inconsistent with the small

⁶ See for example P. M. Morse, *Vibration and Sound* (McGraw-Hill Book Company, Inc., New York, 1948), p. 333.

⁷ A. K. Nielsen, Trans. Danish Acad. Tech. Sci. No. 10 (1949).

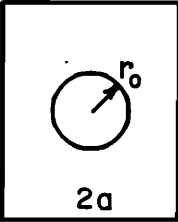
Table I

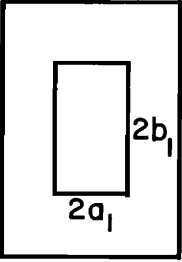
Aperture ↓	Tube ↓	Mass end correction on one side of the aperture. ↓
Circular	Circular	



$$\delta = \sqrt{A_0} \frac{4}{\sqrt{\pi}} \frac{1}{\xi} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{J_1^2(q_{mn}\xi) J_m^2(q_{mn}a/R)}{q_{mn}^3 (1 - m^2/q_{mn}^2) J_m^2(q_m)}$$

$$\xi = r_0/R \quad A_0 = \pi r_0^2$$

Circular	Rectangular	
		$\delta = \sqrt{A_0} \frac{4}{\pi^{3/2}} (\xi \eta)^{-1/2} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} v_{mn} \frac{J_1^2(\pi \sqrt{m^2 \xi^2 + n^2 \eta^2})}{[m^2 b/a + n^2 a/b]^{3/2}}$ $v_{0n} = 1/2 \quad v_{mn} = 1$ $A_0 = \pi r_0^2 \quad \xi = r_0/a \quad \eta = r_0/b$

Rectangular	Rectangular	
		$\delta = \sqrt{A_0} \frac{2}{\pi} \sum_m \sum_n v_{mn} \frac{G_{mn}}{[(b_1/a)m^2 + (a_1/b)n^2]^{1/2}}$ $G_{mn} = \left[\frac{\sin(\pi m \xi)}{\pi m \xi} \frac{\sin(\pi n \eta)}{\pi n \eta} \right]^2 \xi \eta$ $v_{00} = 0 \quad v_{0n} = v_{m0} = 1/2 \quad v_{mn} = 1$ $\xi = a_1/a \quad \eta = b_1/b \quad A_0 = 4a_1 b_1$

signal assumption on which the linear wave equation is based. A thorough analysis should therefore include the nonlinear terms in the equations.

Mass End Correction for Various Apertures

Assuming a uniform velocity distribution, we have calculated the interior end correction δ_i for a few different aperture and cavity geometries of practical importance and the results are listed in Table I. The calculations are presented in Appendix I. The end corrections in three of the simplest cases, namely, a concentric aperture in a circular tube, a circular aperture in a square tube, and a square aperture in a square tube, are plotted in Fig. 3 as a function of ξ , defined in Table I.

Since $\delta_i(A)^{1/2}$ can be considered the same for these three cases and practically linearly dependent on ξ

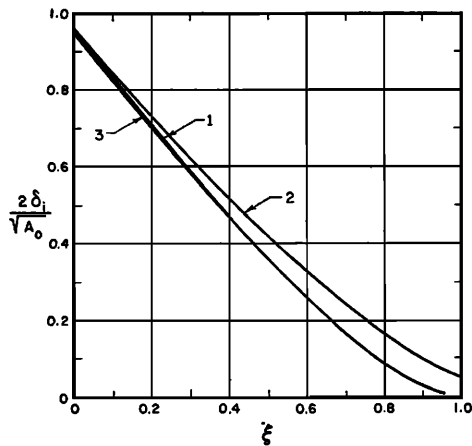


FIG. 3. The mass end correction of concentric circular and square apertures in a tube. (1) Circular aperture in a circular tube. (2) Circular aperture in a square tube. (3) Square aperture in a square tube.

for small values of ξ , we may introduce an approximate expression for the end correction for these apertures. It follows from Fig. 3 that for $\xi < 0.4$, the end correction δ_i can be written approximately

$$\delta_i \approx 0.48(A)^{1/2}(1 - 1.25\xi), \quad (4)$$

an expression that has proved to be in very good agreement with measurements.

For a circular aperture in an infinite wall, it is well known that $\delta/(A_0)^{1/2}$ equals $(8/3)\pi^{-1} \approx 0.48$. In Fig. 3 we see that the end correction for the circular aperture in a cylindrical tube approaches this same value when ξ goes to zero.

When ξ approaches unity the end correction goes to zero as expected, except for the circular aperture in a square tube. In that case there is a little "space" left for higher-order modes to develop in the cavity around

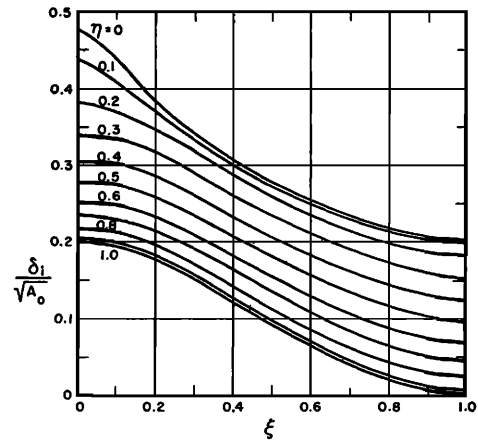


FIG. 4. Mass end correction on one side of a rectangular aperture in a square tube. For explanation of the parameters ξ and η see Table I.

the aperture even if the diameter of the aperture equals the side of the tube.

The end correction for a rectangular aperture in a cavity with a square cross section, obtained by setting $a=b$ in Eq. (4), is shown in Fig. 4. These curves are useful in connection with the design of resonators or facings with slits used to some extent in architectural acoustics.

Of special interest is the end correction for the eccentric circular aperture as given in Table I. The derivation of this expression is shown in Appendix II. The end correction is computed as a function of a , i.e., the distance between the aperture centrum and the tube axis for a single value of $\xi = r_0/R$, namely, $\xi = 0.15$. The result is shown in Fig. 6 where the curve is labeled $\delta_{11}/(A_0)^{1/2}$. The interesting feature of this result is that the end correction is a minimum, or the conductivity of the aperture a maximum when a is approximately equal to 0.5, that is, when the aperture is located half-way between the axis and the wall of the tube. The curve indicates a maximum value of the end correction when $a=R$ but since $r_0=0.15R$ in this case it follows that a cannot be larger than 0.85 and the maximum end correction is therefore obtained for $a=0$.

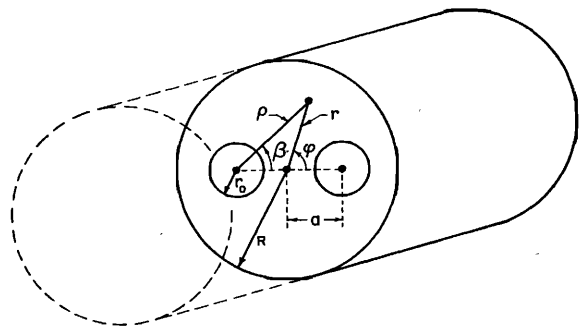


FIG. 5. This figure refers to the calculation of the interaction impedance between two eccentric circular apertures in a tube.

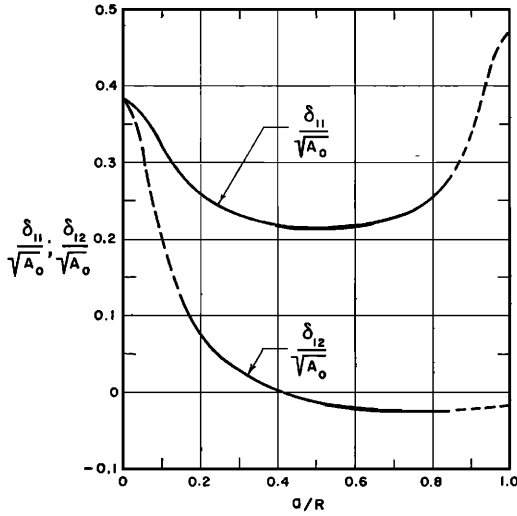


FIG. 6. Mass end correction δ_{11} of an eccentric circular aperture in a circular tube as shown in Table I, and the interaction end correction δ_{12} between two apertures as shown in Fig. 5 for the special value $r_0/R=0.15$.

Interaction between Two Circular Apertures

Again considering a cavity with a circular cylindrical cross section, it is of interest to study the resulting end correction of two circular apertures in the tube as shown in Fig. 5. If interaction is disregarded, each aperture separately should have an end correction approximately $\delta_i = 0.48(A_0)^{1/2}$. Let us for the purpose of illustration consider the aperture in a thin infinite wall for which the total specific acoustic mass reactance of the aperture would be $X = \omega\rho(\delta_i + \delta_e)/A_0 = \omega\rho 2\delta_i/A_0 \approx 0.96\omega\rho/(A_0)^{1/2}$. The effect of interaction can be demonstrated in a simple way by considering two apertures in a wall, first very far apart from each other and then very close to each other. In the first case the combined acoustic reactance of the two supposedly independent apertures in parallel is $X/2 = 0.48\omega\rho/(A_0)^{1/2}$ and in the second case the two apertures combined into one give a mass reactance equal to $X' = 0.96\omega\rho/(2A)^{1/2} = \sqrt{2} \cdot 0.48\omega\rho/(A_0)^{1/2}$. In other words, the combined mass reactance when the holes are set together is higher than when they are separate; in fact, it is higher by a factor of $\sqrt{2}$. The same qualitative argument can be applied also in the case illustrated in Fig. 5, where the two apertures are in a thin wall in a tube which serves to demonstrate the resonator case. Also here it is expected that the combined mass reactance for the two holes very close together should be approximately $\sqrt{2}$ times larger than when the holes are far apart.

To study the interaction effect as a function of distance between the holes, we calculate the pressure p_{12} caused by the first "aperture piston" at the surface of the second and determine the interaction specific impedance

$$z_{12} = \frac{1}{UA_2} \int_{A_2} p_{12} dA_2, \quad (5)$$

where U is the particle velocity in the second aperture. Because of symmetry the particle velocities in both apertures will be the same in this case. The apertures are both circular with a radius r_0 and located at a distance a from the axis of a tube of radius R .

When calculating p_{12} the results from the analysis of the eccentric aperture can be applied and the evaluation of the integral can be done with the help of the addition theorem for the Bessel functions. The details of the calculations are presented in Appendix II. It is shown there that the end correction $\delta_{12} = z_{12}/(-i\omega\rho)$, corresponding to the interaction impedance z_{12} , can be written

$$\delta_{12} = (A)^{1/2} \frac{4}{(\pi)^{1/2} r_0} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} (-1)^m F_{mn}, \quad (6)$$

where

$$F_{mn} = \frac{[J_1(b_{mn}r_0)J_0(b_{mn}a)/J_1(b_{mn}R)]^2}{(b_{mn}R)^3[1 - (m/b_{mn}R)^2]},$$

and $b_{mn}R$ is the n th root to the equation

$$J_m'(b_{mn}R) = 0.$$

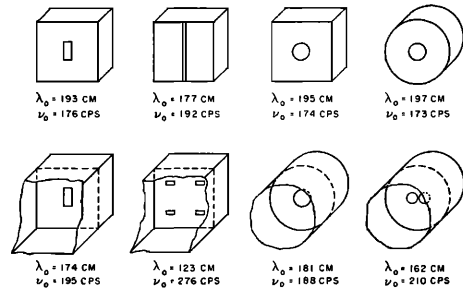


FIG. 7. Examples of resonators with the same volume and aperture area but with different resonance frequencies. Volume of resonator = 5000 cm³. Front area 20 × 20 cm². Total aperture area = 12 cm².

It should be noted that the expression for the interaction end correction δ_{12} has the same form as the end correction δ_{11} for the single eccentric aperture,

$$\delta_{11} = (A)^{1/2} \frac{4}{(\pi)^{1/2} r_0} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} F_{mn}, \quad (7)$$

except for a factor $(-1)^m$.

The interaction end correction δ_{12} has been computed as a function of a for the particular value of $r_0/R=0.15$, and the result is shown in Fig. 6, where the end correction δ_{11} is also shown. The part of the curve outside the range $0.15 < a/R < 0.85$ has, of course, no meaning in this case and is shown dashed.

The largest value of the total combined end correction equals $0.42(A)^{1/2}$ and is obtained when the two apertures just touch each other, i.e., when $a/R=0.15$. The minimum value of the same quantity is $0.20(A)^{1/2}$ corresponding to $a/R \approx 0.6$. The maximum combined total acoustic reactance of the two apertures is thus

$0.84\omega\rho/2(A)^{\frac{1}{2}}$ and the minimum value is $0.40\omega\rho/2(A)^{\frac{1}{2}}$. For comparison it should be noted that the acoustic reactance of a concentric circular aperture with the area $2A$ equals $1.0\omega\rho/2(A)^{\frac{1}{2}}$ according to the curve in Fig. 4.

It is therefore clear that in order to obtain maximum acoustic reactance of a given open area in a partition one should concentrate the open area into one single aperture in the center of the tube. This is important to realize in the design of resonators and acoustic filters.

Examples

In order to calculate the resonance frequency we need not only the interior end correction but also the exterior one. If the resonator is in a plane wall, the exterior end correction can be found from the curves in Figs. 3 and 4 by setting $\xi=0$. As an example let us consider a rectangular box resonator with square cross section with the side 20 cm and a volume 5000 cm³. The aperture is rectangular with the dimensions $a_1=3$ cm and $b_1=4$ cm, and is located in the center of the 20-cm by 20-cm side wall, as shown schematically in Fig. 7. The side wall coincides with the infinite wall in

TABLE II. Measurements of aperture reactance.

Frequency ν cps	Aperture diameter $2r_0$ cm	Measured specific reactance per unit length of the aperture	Calculated acoustic reactance per unit length $\rho\omega \left(1 + \frac{\delta_v}{r_0}\right)$
234	0.5	1.78	1.85
234	1.0	1.79	1.81
246	1.4	1.83	1.90
246	2.0	1.88	1.89

which the resonator is placed. The internal end correction is found from Fig. 4 in which we now have $\xi=3/20=0.15$ and $\eta=6/20=0.3$. The corresponding interior end correction thus becomes $\delta_i \approx 0.36(A)^{\frac{1}{2}}$, the exterior correction $\delta_e \approx 0.48(A)^{\frac{1}{2}}$, and hence the total $\delta=0.84(A)^{\frac{1}{2}}$. Assuming the thickness of the wall negligible, the resonance wavelength becomes 219 cm. Correspondingly, the resonance wavelength for a few resonators with the same volume and the same aperture area as before but with different shape have been calculated and listed in Fig. 7. In the second of these examples the aperture is in the form of a vertical slit extending over the entire front surface. The interior end correction found from Fig. 4 in which $\xi=0.045$ and $\eta=1$ is then $\delta_i=0.19(A)^{\frac{1}{2}}$, and the exterior is taken to be $\delta_e=0.48(A)^{\frac{1}{2}}$.

When the resonators are placed at the end of a tube, as shown in the last four examples in Fig. 7, the exterior end correction equals the interior and the resonance frequency will be higher than for the corresponding resonators in an infinite wall. When the total aperture area is split up between several apertures, the resonance frequency will in general be considerably increased. In the sixth resonator in Fig. 7, the aperture is split up

into four smaller apertures. By symmetry the resonance frequency in this case will be the same as for one of the elementary four resonators that make up the original resonator.

In the last example in Fig. 7 the study of the interaction between two apertures in a circular cylindrical tube has been applied and the resonance frequency in this case is found by the aid of Fig. 6 in which we have to use $a/R=0.15$.

Reactance Measurements

The discussion so far has been essentially limited to thin apertures. Assuming that the motion in an aperture of finite thickness is the same as in a tube of infinite length, the additional mass reactance due to the thickness of the aperture should be⁸

$$X_n = \rho c k t (1 + \delta_v/r_0),$$

where

$$\delta_v = \left(\frac{2\mu}{\omega\rho} \right)^{\frac{1}{2}} \quad (\mu = \text{coefficient of viscosity of air } k = \omega/c),$$

is the viscous boundary layer thickness.

To check this expression experimentally, a series of measurements was performed on a set of circular apertures with diameters ranging from 0.36 to 2.0 cm and thicknesses from 0.05 cm to 2.5 cm. The frequency range was 200 to 1000 cps.

The conventional standing-wave method with a traveling microphone was used. The plate containing the aperture was placed at the end of the impedance

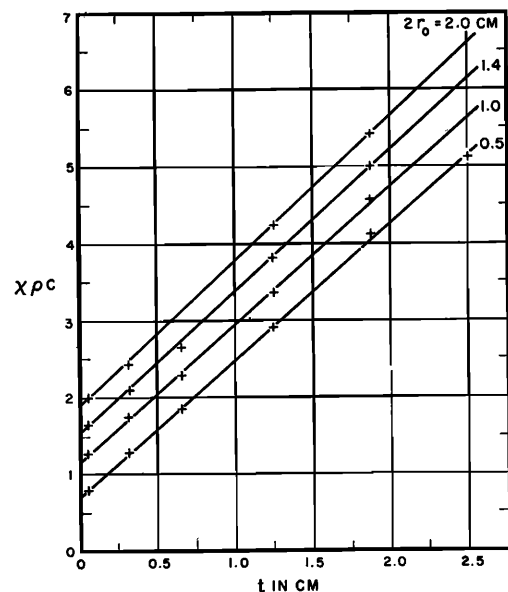


FIG. 8. Measured specific reactance of circular apertures as a function of plate thickness t for different aperture diameters d .

⁸ Crandall, *Theory of Vibrating Systems and Sound* (D. Van Nostrand Company, Inc., New York, 1927), p. 229.

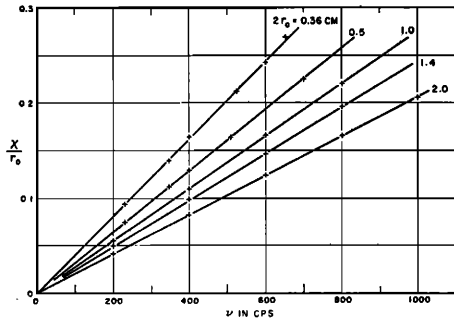


FIG. 9. Measured mass reactance of circular apertures in a tube as a function of frequency. Aperture thickness $t=0.05$ cm.

tube and was backed by a tube section terminated by a rigid piston. The length of this section was always chosen to be a quarter wavelength so that the plane-wave reactance of the cavity behind the aperture was zero. The measured impedance was then just the aperture impedance.

Curves illustrating the measured specific reactances as a function of aperture thickness t are shown in Fig. 8. The reactance increases linearly with t and the slope should be ω . The maximum deviation from this value is less than 4 percent as can be seen in Table II.

The residual part of the reactance at $t=0$, shown in Fig. 8, represents the end correction. A special study of this part of the reactance was made by measurements on a group of very thin apertures, $t=0.05$ cm, for which the end correction is the essential part of the mass. The results of these measurements are shown in Fig. 9 where the quantity X/r_0 is plotted as a function of frequency, X being the specific reactance of the aperture. The experimental values lie essentially on straight lines, indicating that the kinetic mass is independent of frequency in the frequency range under consideration. It may be mentioned that a plot of $(X_0/r_0)=(16/3\pi)\omega\rho$, representing the reactance for an aperture in an infinite wall, is independent of orifice diameter, which is evidently not true in the tube case.

From the slope S of a reactance curve in Fig. 9, we get the total kinetic mass of the aperture $m_{\text{tot}}=(r_0 S/2\pi)$, which expressed as an equivalent neck length is $l=(S/2\pi\rho)r_0$. By subtracting the neck length t we obtain the total end correction $2\delta=l-t$, which is listed in Table III together with the values calculated from the equations in Table I. The agreement between theoretical and experimental values is evidently very good, the average percentage error for the five apertures being 2.6 percent.

The measured values are consistently somewhat smaller than the calculated. This is expected from the fact that any assumed velocity distribution different from the correct one gives an end correction which is always too big. However, the deviations found here are so small that the plane velocity distribution can be considered to yield the mass reactance of an aperture

with an accuracy which is within the experimental error of most acoustic measurements.

IV. THE DISSIPATION IN THE RESONATOR

If no damping material in the form of screens or porous material is introduced into the resonator, the acoustic dissipation is due mainly to viscosity and heat conduction losses on the surfaces of the resonator. In addition, nonlinear losses caused by circulation effects and turbulence come in at sufficiently high sound intensities. We shall return to these nonlinear effects later and consider now only the "linear" losses.

Effect of Viscosity

The dissipation caused by viscosity can be calculated approximately from the integral⁹

$$W_v = \frac{1}{2} \int_S R_s |U_s|^2 dS, \quad (8)$$

where U_s is the tangential velocity amplitude at the surface S calculated from the wave equation neglecting

TABLE III. Measurements of end correction.

Diameter $2r_0$ cm	Measured slope S of curves in Fig. 9	Total equivalent neck length l cm	Neck length t cm	End correction $2\delta=l-t$	2δ calculated
0.36	$0.406 \cdot 10^{-3}$	0.390	0.097	0.283	0.284
0.50	$0.324 \cdot 10^{-3}$	0.430	0.051	0.379	0.387
1.00	$0.276 \cdot 10^{-3}$	0.736	0.051	0.685	0.707
1.40	$0.245 \cdot 10^{-3}$	0.920	0.051	0.869	0.890
2.00	$0.207 \cdot 10^{-3}$	1.110	0.051	1.059	1.12

viscosity, and R_s is a "surface resistance." If the radius of curvature of the surface under consideration is large compared to the viscous boundary layer thickness, one can as a good approximation use the resistance calculated from the problem of an oscillatory flow over an infinite plane surface. This resistance is¹⁰

$$R_s = \frac{1}{2} (2\mu\rho\omega)^{\frac{1}{2}}, \quad (9)$$

where μ is the viscosity and ρ the density of the medium. At ordinary room temperature, $t \approx 20^\circ\text{C}$, the numerical value for the surface resistance becomes

$$R_s \approx 0.83 \cdot 10^{-3} (\nu)^{\frac{1}{2}}, \quad (10)$$

where ν is the frequency.

For surfaces with small radii of curvature, Eq. (10) may be in serious error. In the case of an aperture in a thin plate, for example, the radius of curvature on the rim of the aperture may be very small, of the same order of magnitude as the boundary layer thickness

⁹ L. Cremer, Arch. elektr. Übertragung 2, 136 (1948), and reference 7.

¹⁰ Reference 1, Vol. II, p. 318.

$\delta = (\mu/\omega)^{1/2}$. Since the tangential particle velocity is very large in this region, a considerable part of the dissipation may actually take place here, and it can be shown that the use of Eq. (10) as the surface resistance in such a case will result in a too small dissipation.^{11†}

The viscous dissipation on the aperture plate can be considered made up of two parts, namely, the losses on the side walls in the aperture neck and the losses on the parallel wall surfaces of the plate. The former contributions equal $\frac{1}{2}R_S 2\pi r_0 t U_0^2$ and the latter is calculated from the integral $\frac{1}{2}\int R_S |u_r|^2 dS$ where u_r is the tangential velocity over the parallel plate surfaces. If the velocity in the aperture is assumed uniform, Nielsen has shown⁷ that this integral can be expressed simply as

$$\frac{1}{2}\int |u_r|^2 R_S dS = \frac{1}{2}R_S \pi r_0^2 U_0^2,$$

where $r_0/R \ll 1$.

The total dissipation in and around the aperture is, therefore, $\frac{1}{2}R_S U_0^2 [2t/r_0 + 2]$, and the corresponding

TABLE IV. Measurements of neck resistance.

Frequency cps	Aperture diameter $d = 2r_0$ cm	Measured slope $S = 2R_s/r_0$ (see Fig. 10)	Surface resistance R_s obtained from measurements	Calculated surface resistance from Eq. (11a)
229	0.5	0.105	0.013(1)	0.0127
234	1.0	0.0570	0.014(1)	0.0129
234	1.4	0.0378	0.013(2)	0.0129

acoustic specific aperture resistance becomes

$$\theta_i = \frac{R_S}{\rho c} \left(\frac{2t}{r_0} + 2 \right) = \frac{2R_S}{\rho c} \frac{1}{r_0} (t + r_0) = \theta_n + \theta_e, \quad (11)$$

$$\theta_n = \frac{2R_S}{\rho c} \frac{t}{r_0}; \quad \theta_e = \frac{2R_S}{\rho c}. \quad (11a)$$

The dissipation on the parallel walls of the aperture can thus be interpreted as analogous to a resistance end correction $\Delta l = r_0$.

If, instead of using a uniform velocity in the aperture, we choose the incompressible nonviscous flow distribution with a velocity going to infinity at the sharp edges, the integral (8) diverges. This indicates that the true value for the viscous end correction is larger than the value found above. This conclusion is also confirmed by the measurements reported briefly below. The viscous end correction thus obtained experimentally was in all cases found to be close to $2r_0$ rather than the

¹¹ A. Kuckes and U. Ingard, J. Acoust. Soc. Am. **25**, 798 (1953). See also J. Milton Andres, Ph.D. dissertation, p. 73 (Physics Department, Massachusetts Institute of Technology, 1953).

† Hydrodynamically speaking, one can say that Eq. (10) and the corresponding integral (8) are valid for very large Reynolds numbers.

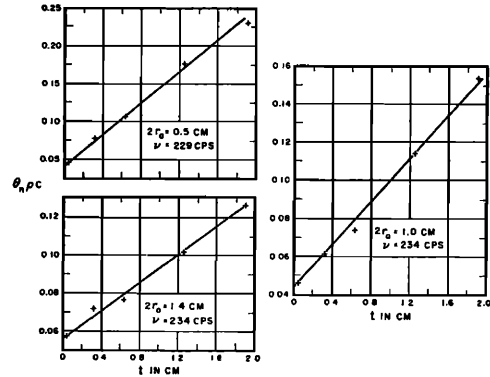


FIG. 10. Measured values of the resistance of a circular aperture as a function of aperture thickness.

value r_0 obtained from the theory based on uniform velocity distribution.

Effect of Heat Conduction

The dissipation in a resonator due to heat conduction can be calculated approximately from the integral^{7,9}

$$W_h = \frac{1}{2} \int H_s |P|^2 dS, \quad (12)$$

where

$$H_s = (\kappa - 1) \frac{k}{\rho c} \left(\frac{\lambda_h}{2\omega c_p \rho} \right)^{1/2},$$

λ_h = coefficient of heat conduction of air,

c_p = specific heat at constant pressure of air,

ρ = density of air, $\kappa = c_p/c_v$,

$|P|$ = magnitude of sound pressure at surface.

Under normal atmospheric conditions (1 atmos $T \approx 20^\circ\text{C}$), the numerical value of H_s is

$$H_s \approx \frac{1}{\rho c} 10^{-6} (\nu)^{1/2} \quad (\nu = \text{frequency}). \quad (13)$$

It can be shown that in cases of practical importance the heat conduction losses are negligible compared with the viscous losses. As pointed out by Nielsen,⁷ this fact seems to have been overlooked by several investigators who have applied the Kirchhoff formula for the attenuation of plane waves indiscriminately for calculation of the dissipation in the neck of a resonator. This is not possible since in the neck of a resonator the relation between pressure and velocity is not the same as in an infinitely long tube. The velocity plays a much more important part than the pressure in the resonator neck as far as dissipation is concerned.

Measurements of Aperture Resistance

In order to separate the neck dissipation from the end-correction dissipation experimentally, the resistance of the aperture was measured as a function of neck length l . The measurements were made in the same way as the reactance measurements described above.

The resistance thus found is shown in Fig. 10 to be essentially linearly increasing with the neck length. From the slope of this curve the resistance per unit length of the neck is found to be in fairly good agreement with the value obtained from Eq. (11a). The maximum deviation between the measured and calculated values is 9 percent in measurements on three apertures with diameters 0.5, 1.0, and 1.4 as seen in Table IV.

The residual part of the measured resistance that remains when $l=0$, as seen in Fig. 10, corresponds to the resistance end correction Δ_l . The experimental value of Δ_l thus found is considerably larger than the value calculated from Eq. (11a). In fact, the measured values of the end correction are just about twice the value calculated from Eq. (11a). A discrepancy is expected not only on the basis already discussed concerning the choice of velocity distribution in the aperture. Also the use of the expression for the surface resistance given in Eq. (10) is not justified in the region about the rim of the aperture where the effective radius of curvature may be of the same order of magnitude as the boundary layer thickness. However, a rigorous solution of the problem is out of reach at present, and an analysis of the type presented may be subject to modification by experimental findings. In what follows we shall thus use the experimentally found value for the resistance end correction

$$\Delta_l \approx 2r_0.$$

Corresponding calculations and measurements on a spherical resonator in a free field have shown that the resistance also in that case can be calculated in this same way, and it is reasonable to assume that it can be used approximately for any resonator of the Helmholtz type.

V. DESIGN CONSIDERATIONS

Once the mass reactance and the resistance have been expressed in terms of the geometrical parameters of the resonator, we are able to investigate the question of optimum design for maximum absorption. Let us start these considerations by a study of the absorption and scattering from a spherical resonator in a free field. The resonator consists of a spherical shell of inner diameter $2a$ and with a circular aperture with radius r_0 . The thickness of the shell at the aperture is t .§

§ In the case of a nonspherical resonator the quantity $2a$ represents the inner diameter of an equivalent sphere with the same volume as the resonator.

Resonator Parameters

To describe the resonator acoustically we shall use the dimensionless parameters r_0/a , t/r_0 as independent variables. The expression for the resonance frequency ν_0 can then be written in the form

$$k_0 a = \left(\frac{3}{4} \frac{r_0}{a} \right)^{\frac{1}{2}} \left(\frac{\delta}{r_0} + \frac{t}{r_0} \right)^{-\frac{1}{2}} \quad (14)$$

or, introducing $\delta_0 = 1.70 r_0$, i.e., the end correction for an aperture in an infinite screen,

$$k_0 a = 0.666 (r_0/a)^{\frac{1}{2}} (\beta + 0.59t/r_0)^{-\frac{1}{2}}. \quad (15)$$

The ratio $\beta = \delta/\delta_0$ is close to unity for small apertures. A good average value is $\beta \approx 0.9$. For greater accuracy the curves in Figs. 3 and 4 should be consulted. The quantity β varies relatively slowly with r_0/a and will be considered as a constant in this analysis.

The expression for the aperture resistance can be written, using the resistance end correction $\Delta_l = 2r_0$,

$$\theta_i = \frac{4R_s}{\rho c} \left(1 + 0.5 \frac{t}{r_0} \right). \quad (16)$$

In order to include the possibility of having an additional resistance $\rho c \theta_a$ in the aperture, for example, in the form of a cloth, screen, or some other thin porous layer, we write the total internal resistance as

$$\theta_i' = \frac{4R_s}{\rho c} \left(\epsilon + 0.5 \frac{t}{r_0} \right), \quad (17)$$

where

$$\epsilon = 1 + \rho c \theta_a / 4R_s.$$

In most cases it can be assumed that θ_a has the same frequency dependence as R_s , i.e., proportional to $\nu^{\frac{1}{2}}$, and ϵ can thus be considered to be independent of frequency.

The specific radiation resistance for the aperture in the resonator equals $(\frac{1}{4}) (kr_0)^2$ and the total radiation resistance at resonance can be written, using Eq. (15),

$$\theta = 4(R_s/\rho c)(\epsilon + 0.5t/r_0)[1 + C(k_0 a)^6], \quad (18)$$

where

$$C = 0.316(R_s/\rho c)^{-1} h(t/r_0).$$

The hole parameter

$$h(t/r_0) = \frac{(\beta + 0.59t/r_0)^2}{(\epsilon + 0.5t/r_0)}, \quad (19)$$

is plotted in Fig. 11 as a function of ϵ for different values of t/r_0 . This parameter h is practically equal to unity for small values of t/r_0 when no extra resistance is introduced, i.e., when $\epsilon = 1$. In plotting the graphs in Fig. 11 an average value of β has been used, $\beta = 0.9$.

The expression for the absorption cross section at

resonance given in Eq. (3) can now be written

$$\sigma_a = \frac{\lambda_0^2}{4\pi} \frac{4C(k_0a)^6}{[1+C(k_0a)^6]^2} F, \quad (20)$$

$$F = 1 + \frac{9}{16} (k_0a)^2 (1 + \cos\theta_0)^2,$$

$$\sin\theta_0 = r_0/a.$$

The factor F which in the limiting case of $\omega=0$ equals unity has been determined here to the second power in ka from a detailed study of the scattering from a spherical resonator which will not be presented here. Since the second term in F is a small correction term we can consider F to be a constant in the analysis that follows.

The scattering cross section $\sigma_s = (\theta_r/\theta_i)\sigma_a$ can be expressed similarly. Starting from the definition of the Q value of the resonator $Q = \omega_0 m / \theta \rho c$ and introducing $m = \rho 1.7 r_0 (\beta + 0.59 l / r_0)$ and the resistance given by Eq. (18), we find

$$Q = 3.03 \frac{C(k_0a)^3}{1 + C(k_0a)^6}. \quad (21)$$

Correspondingly, from the exponential decay of the pressure in the resonator, determined by the factor $\exp[-(\omega_0 \tau / 2)(1/Q)]$, the reverberation time of the resonator becomes

$$T = 13.8 \frac{Q}{\omega_0} = 41.8 \frac{1}{\omega_0} \frac{C(k_0a)^3}{1 + C(k_0a)^6}. \quad (22)$$

It is clear from the expressions for σ_a , Q , and T that these quantities will have maximum values for certain values of the resonator parameters. In finding these optimum values it should be remembered that C is a function of frequency since $R_s \sim (\nu)^{1/2}$. In other words σ_a , Q , and T are not strictly functions of (k_0a) only. Hence, one has to consider separately the two cases when a and k_0 are considered as variables. If we, for example, assume the radius of the sphere given and vary the frequency, we should set $K = C(k_0a)^{1/2}$ and study the function

$$\sigma_a = 4\pi a^2 \frac{K(k_0a)^{7/2}}{[1 + K(k_0a)^{11/12}]^2} F, \quad (23)$$

with k_0 as variable. In cases of more practical importance, however, the resonator frequency is given, and we ask for the radius a (or the volume) that makes any of the quantities mentioned a maximum. Then the expressions (20), (21), and (22) can be used as they stand since C is now strictly a constant.

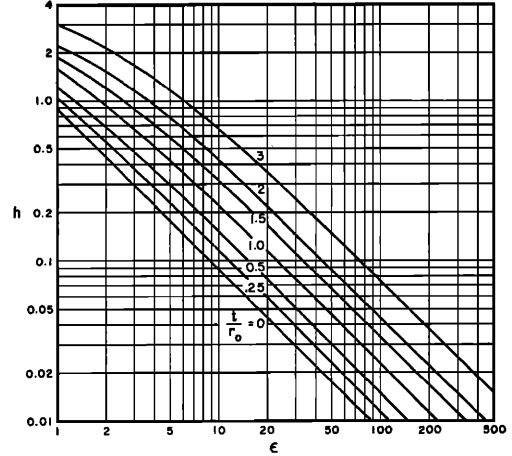


FIG. 11. The aperture parameter h given in Eq. (19) presented as a function of ϵ with l/r_0 as parameter.

Optimum Design for Maximum Absorption Cross Section for a Given Resonance Frequency and Ratio l/r_0

If we by a_1 denote the value of the sphere radius at which the absorption cross section is a maximum we find||

$$k_0 a_1 = C^{-1/6} = 1.08 (2R_s / \rho c)^{1/6} h^{-1/6}, \quad (24)$$

(0.965)

with the corresponding value for the absorption cross section

$$\sigma_{a1} = \frac{\lambda_0^2}{4\pi} F. \quad (25)$$

Numerically, the value for the optimum value for the sphere radius becomes

$$a_1 = 1080 \nu_0^{-11/12} h^{-1/6} \text{ cm (if } \nu_0 \text{ in cps).} \quad (26)$$

If, for example, $\nu_0 = 200$ cps and $l \ll r_0$, we get $h \approx 0.9$ and $a_1 = 8.4$ cm.

Using the quantities a_1 and σ_{a1} , the absorption cross section at resonance for an arbitrary value of a can be written in the form

$$\frac{\sigma_a}{\sigma_{a1}} = 4 \frac{(a/a_1)^6}{[1 + (a/a_1)^6]^2}. \quad (27)$$

Correspondingly, we find a maximum value of Q ,

$$Q_1 = 1.20 (R_s / \rho c)^{-1/2} h^{1/2} = 189 \nu_0^{-1/4} h^{1/2}, \quad (28)$$

(0.85)

when

$$a = a_1.$$

The normalized expression for Q is

$$\frac{Q}{Q_1} = 2 \frac{(a/a_1)^3}{1 + (a/a_1)^6}. \quad (29)$$

|| The numbers within parenthesis refer to a resonator in an infinite wall.

Correspondingly, for the reverberation time,

$$T_1 = 2.64 (R_s / \rho c)^{-1/2} \nu_0^{-1} h^{1/2} = 415 \nu_0^{-5/4} h^{1/2} \text{ sec}, \quad (30)$$

(293)

when

$$a = a_1.$$

The normalized form is the same as for Q , i.e.,

$$\frac{T}{T_1} = 2 \frac{(a/a_1)^3}{1 + (a/a_1)^6}. \quad (31)$$

Optimum Design When Resonator Volume and t/r_0 Are Given

If we consider the volume of the resonator constant and ask for the frequency or the area of the aperture

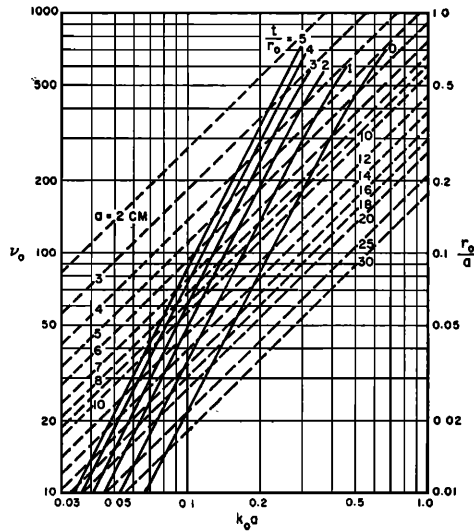


FIG. 12. Resonance frequency chart representing Eq. (15). To find the resonance frequency ν_0 determine the intersection between the line for a given t/r_0 and the horizontal line representing a given r_0/a . From this intersection go in vertical direction to the intersection with the a line and read off the corresponding resonance frequency (and $k_0 a$ value). t = aperture thickness, r_0 = aperture radius, a = resonator radius.

(for constant t/r_0) at which σ_a , Q , and T are maximum, we have to use Eq. (23) and corresponding equations for Q and T . The results of the calculations are summarized below.

The optimum values of k_0 and r_0 corresponding to maximum resonance absorption are

$$k_{01} a = \left(\frac{7}{15} \frac{1}{K} \right)^{2/11} = \frac{0.948 \theta_s^{2/11} h^{-2/11}}{(0.835)}, \quad (32)$$

where

$$\theta_s = (2R_s / \rho c) (k_0 a)^{-1/2},$$

and

$$\begin{aligned} r_{01} &= 2.03 \theta_s^{4/11} (\beta + 0.59 t/r_0), \\ &= 0.242 a^{9/11} h^{-4/11} (\beta + 0.59 t/r_0). \end{aligned} \quad (33)$$

The maximum absorption cross section is

$$\begin{aligned} \sigma_{a1} &= 1.15 \pi a^2 K^{4/11} F = 0.845 \pi a^2 \theta_s^{-4/11} h^{4/11} F \\ &= 7.1 \pi a^2 a^{2/11} h^{4/11} F, \end{aligned} \quad (34)$$

and the normalized expression for σ_a

$$\begin{aligned} \frac{\sigma_a}{\sigma_{a1}} &= \left(\frac{22}{15} \right)^2 \frac{(r_0/r_{01})^{7/4}}{\left[1 + \frac{7}{15} (r_0/r_{01})^{11/4} \right]^2} \\ &= \left(\frac{22}{15} \right)^2 \frac{(\nu_0/\nu_{01})^{7/2}}{\left[1 + \frac{7}{15} (\nu_0/\nu_{01})^{11/2} \right]^2}. \end{aligned} \quad (35)$$

Corresponding results for the Q value are

$$Q = 3.03 K \frac{(k_0 a)^{5/2}}{1 + K (k_0 a)^{11/2}}, \quad (36)$$

$$k_{01} a = \left(\frac{5}{6} \frac{1}{K} \right)^{2/11} = 1.05 \theta_s^{2/11} h^{-2/11} = 0.365 a^{-1/11} h^{-2/11}, \quad (37)$$

$$\begin{aligned} r_{01} &= 2.49 \theta_s^{4/11} a h^{4/11} (\beta + 0.59 t/r_0) \\ &= 0.298 a^{4/11} h^{-4/11} (\beta + 0.59 t/r_0), \end{aligned} \quad (38)$$

$$Q_1 = 1.19 \theta_s^{-6/11} h^{6/11} = 25 V^{1/11} h^{6/11}, \quad (V = \text{volume}) \quad (39)$$

$$\frac{Q}{Q_1} = \frac{5}{11} \frac{(\nu_0/\nu_{01})^{5/2}}{1 + \frac{5}{6} (\nu_0/\nu_{01})^{11/2}} = \frac{5}{11} \frac{(r_0/r_{01})^{5/4}}{1 + \frac{5}{6} (r_0/r_{01})^{11/4}}. \quad (40)$$

And for the reverberation time, we get

$$T = 41.8 \frac{a}{c} \frac{K (k_0 a)^{3/2}}{1 + K (k_0 a)^{11/2}}, \quad (41)$$

$$k_{01} a = \left(\frac{3}{8} \frac{1}{K} \right)^{2/11} = 0.912 \theta_s^{2/11} h^{-2/11} = 0.315 a^{-1/11} h^{-2/11}, \quad (42)$$

$$\begin{aligned} r_{01} &= 1.87 \theta_s^{4/11} a h^{-4/11} (\beta + 0.59 t/r_0) \\ &= 0.150 V^{3/11} h^{-2/11} (\beta + 0.59 t/r_0), \end{aligned} \quad (43)$$

$$T_1 = 16.6 \frac{a}{c} \theta_s^{-8/11} h^{8/11} = 0.0335 a^{15/11} h^{8/11}, \quad (44)$$

$$\frac{T}{T_1} = \frac{11}{8} \frac{(\nu_0/\nu_{01})^{3/2}}{1 + \frac{3}{8} (\nu_0/\nu_{01})^{11/2}} = \frac{11}{8} \frac{(r_0/r_{01})^{3/4}}{1 + \frac{3}{8} (r_0/r_{01})^{11/4}}.$$

There are a few remarks concerning these results which may be worth mentioning. First it should be noted that the maximum absorption cross section in Eq. (34), obtained by varying the resonance frequency and keeping the sphere radius constant, does not occur

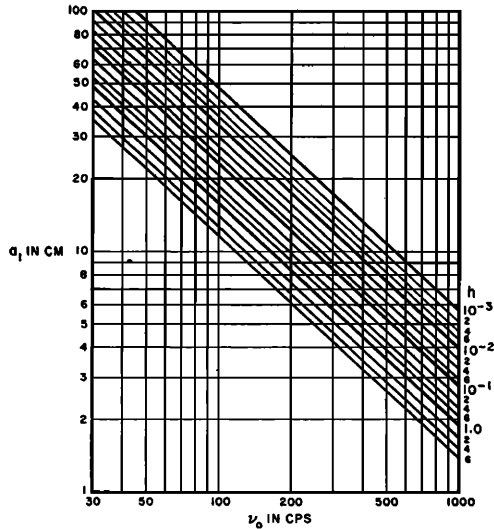


FIG. 13. This figure, based on Eq. (26), gives the optimum resonator radius a for maximum resonance absorption as a function of the resonance frequency ν_0 and the hole parameter h .

when the internal resistance equals the radiation resistance, which is approximately the case described by Eqs. (24)–(31). It is also interesting to realize that the maximum absorption cross section for resonators of “normal” size, with a volume of about 1000 cm^3 is about 11–12 times the geometrical cross section of the resonator. The maximum absorption cross section increases slowly with the volume.

The expression (28) indicates that the Q value should go to infinity when the frequency goes to zero. However, the whole analysis breaks down for very low frequencies since the basic assumption that the boundary layer thickness is small is then violated.

In order to make the analysis complete, we have to investigate the case when both the sphere radius a and k_0 are kept constant and t/r_0 is considered as the variable. It is then seen that in order to obtain an optimum value we must have $dC/d(t/r_0)=0$, or in other words $h'=0$, which leads to the equation

$$(t/r_0)_1 = \frac{0.5\beta - 1.18}{0.59 \cdot 0.5}.$$

However, 0.5β cannot exceed unity and there is no positive value of t/r_0 for which any of the quantities discussed is a maximum or a minimum.

Design Charts and Example

When presenting the results thus found in graphical form, we shall consider only the case when the resonance frequency is given and the sphere radius is varied to make the resonance absorption cross section a maximum. Thus starting with a given resonance frequency and a hole parameter h , we obtain the optimum sphere radius from Eqs. (24) or (26) and the corresponding

maximum values σ_{a1} , Q_1 , and T_1 from the Eqs. (25), (28), and (30). Finally the σ , Q , and T for an arbitrary value of a are found from the normalized expressions (27), (29), and (31). The aperture radius in every case can be obtained from the frequency formula (15). The set of curves representing these equations is presented in Figs. (11)–(16).

As an illustration of the use of the charts let us consider the design of a resonator with a resonance frequency 200 cps and with maximum possible absorption cross section at resonance and with no restrictions on Q and T (which both are maximum when σ_a is a maximum). We assume that there is no additional resistance introduced in the aperture and that $t \ll r_0$, so that the hole parameter is $h \approx 0.9$. From Fig. 13 we obtain the optimum sphere radius $a_1 = 8.4 \text{ cm}$ and the frequency

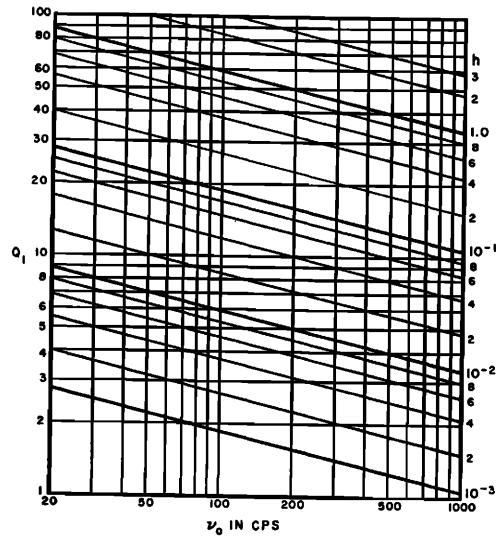


FIG. 14. This figure, based on Eq. (28), gives the maximum possible Q value of a resonator in a free field with a given resonance frequency ν_0 and hole parameter h .

chart in Fig. 12 yields $r_0 = 0.185 = 1.55$. The corresponding maximum absorption cross section at resonance is then $\sigma_{a1} \approx (\lambda_0^2/4\pi)F \approx 1.2\lambda_0^2/4\pi$, or expressed in terms of the cross section of the sphere, $\sigma_{a1} \approx 13.3\pi a^2$. The corresponding Q and T values are obtained directly from Figs. 14 and 15, $Q_1 \approx 50$ and $T_1 \approx 0.56$. It should be pointed out that it is impossible to design a spherical shell Helmholtz resonator in air for a resonance frequency 200 cps with higher values of σ_a , T , and Q than these maximum values.

For practical purposes this optimum design is of no particular value if we are interested in absorption of noise over an extended frequency range. In such a case we shall certainly put a condition on Q to be smaller than a certain value. For the sake of illustrating the design in such a case, let us require $Q_1 < 25$. From Fig. 14 it follows that, in order to have $Q_1 = 25$, the hole parameter must be made $h \approx 0.25$ which, according to Fig. 11, corresponds to an additional resistance in the neck

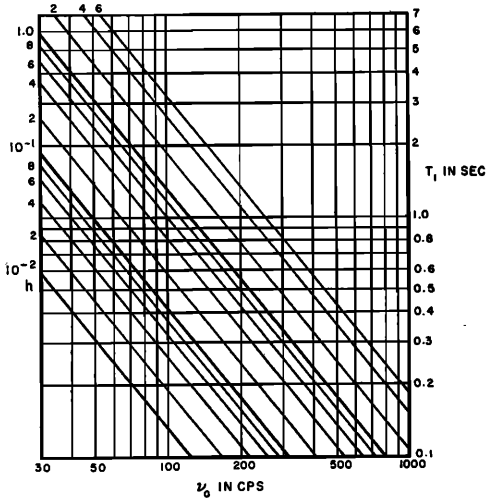


FIG. 15. This figure, based on Eq. (30), gives the maximum possible reverberation time of a resonator in a free field with a given resonance frequency ν_0 and hole parameter h .

$\theta_{\theta} \rho c = (\epsilon - 1)4R_s = (3.5 - 1)4R_s \approx 0.09$ rayl. The corresponding optimum sphere radius is then increased to 10.5 cm (Fig. 13). If we, however, instead of 10.5 choose $a = 8.4$ cm, then $a/a_1 = 0.8$, and we find from Fig. 16 $\sigma_a = 0.65$, $\sigma_{a1} \approx 8.6\pi a^2$ and $Q \approx 0.83$, $Q_1 \approx 20.7$.

The Resonator Array

We have seen that a single resonator in a free field has a maximum resonance absorption cross section approximately equal to $\lambda_0^2/4\pi$. Consider now a single layer of resonators put up in array with a cell area $\lambda_0^2/4\pi$. It may then be tempting to make the erroneous conclusion that since each resonator has an absorption cross section $\lambda_0^2/4\pi$ the incident plane wave in the array should be totally absorbed, and no sound should be transmitted. This is evidently not so.

To investigate this question consider a plane wave of normal incidence on a resonator screen with cell area A . Then due to symmetry, the sound field is analogous to that obtained in the case of a resonator located at the center of a tube with a cross-section area equal to the cell area A_1 of the array. The acoustic impedance of the tube is $Z_0 = \rho c/A_1$ and the acoustic aperture resistance of the resonator is $R_i = \rho c \theta_i/A_0$ where A_0 is the aperture area. The equivalent circuit for this arrangement is then a line of characteristic impedance Z_0 terminated by the resonator impedance in parallel with the impedance Z_0 . The inductance corresponding to the constriction between the tube and the resonator is generally small and may be neglected here.

If we set $x = R_i/Z_0$ the power absorbed by the resonator is

$$W_a = 4 \frac{x^2}{(1+x)^2} I_0 A_1, \quad (45)$$

where I_0 is the intensity of the incident wave. Correspondingly the power transmitted is

$$W_t = 4 \frac{x^2}{(1+x)^2} I_0 A_1. \quad (46)$$

To compare this case with the single resonator in a free field, we determine the conditions under which the resonator absorbs maximum power. According to Eq. (45), this occurs when $x = \frac{1}{2}$, and the maximum absorbed power is $I_0 A_1/2$, corresponding to an absorption cross section $A_1/2$. The transmitted power is then $I_0 A_1/4$ and the remaining $I_0 A_1/4$ is reflected. The maximum absorbed power,

$$(W_a)_{\max} = I_0 A_1/2 = (A_0/4\theta_i) I_0,$$

turns out to be the same as for a resonator in a free field. In a free field the scattering cross section was shown to equal the absorption cross section at optimum design. The results obtained here can be interpreted to be analogous to the free-field case; the amount of energy

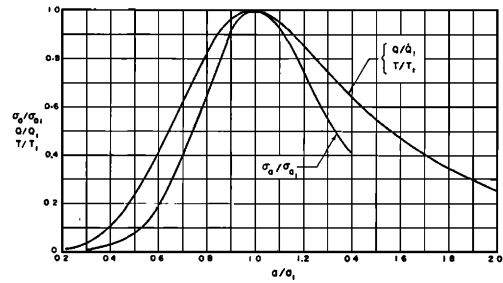


FIG. 16. Normalized curves for absorption cross section σ_a , Q value, and reverberation time T as a function of the resonator radius a . The quantities σ_a , Q , T , and a correspond to optimum design.

absorbed is $I_0 A_1/2$ and an equal amount is scattered, half of it in the forward direction and the other half in the backward direction. Since the transmitted or forward scattered power is $I_0 A_1/4$, the transmission loss of the resonator array for this design is 6 db. A higher value of this loss can obviously be obtained by making R_i/Z_0 smaller than $\frac{1}{2}$, by moving the resonators closer together. But the absorption *per resonator* is the highest possible when the conditions above are fulfilled. The same results hold clearly also for a resonator built into the side wall of a tube.

In the case of propagation through a three-dimensional array of resonators, the problem is in general quite complicated. However, if the plane wave of sound propagates in a direction normal to a main plane of the rectangular lattice, the problem reduces essentially to that of a periodically loaded one-dimensional line and can be treated according to standard methods.¹²

¹² L. Brillouin, *Wave Propagation in Periodic Structures* (McGraw-Hill Book Company, Inc., New York, 1946).

If the frequency is low enough to make the wavelength much longer than the distance between the resonators, the effect of the lattice can be considered simply as a change in the compressibility of the medium resulting in a complex index of refraction. The compressibility of the medium will be $(1/\rho c^2)n^2$, where the index of refraction $n=n_1+in_2$ corresponds to the propagation constant $k_1=nk$. It can be shown that

$$n_1 = \text{Re}(n) = \left(\frac{1-f}{2} \right)^{\frac{1}{2}} \{ (1+\alpha)^{\frac{1}{2}} + (1-\beta)^{\frac{1}{2}} \}^{\frac{1}{2}},$$

$$n_2 = \text{Im}(n) = \left(\frac{1-f}{2} \right)^{\frac{1}{2}} \{ (1+\alpha)^{\frac{1}{2}} - (1-\beta)^{\frac{1}{2}} \}^{\frac{1}{2}},$$

where

$f = v/V$ the percentage "filling" of the medium,

$$\alpha = \frac{Q^2 f_1}{1 + Q^2 \left(\gamma - \frac{1}{\gamma} \right)^2} \left[(2 + f_1) \frac{1}{\gamma^2} - 2 \right],$$

$$\beta = Q^2 \frac{\left(1 - \frac{1}{\gamma^2} \right)}{1 + Q^2 \left(\gamma - \frac{1}{\gamma} \right)^2} f_1; \quad f_1 = \frac{1}{1-f},$$

$\gamma = \omega/\omega_0$; $\omega_0/2\pi$ = resonance frequency of the resonators,

$Q = Q$ value of resonator.

As an example we have calculated the real and imaginary part of the index of refraction for a resonator

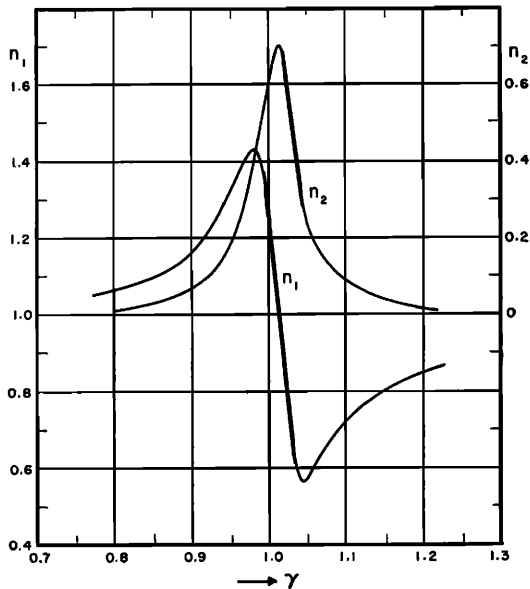


FIG. 17. The real and imaginary parts of the index of refraction $n=n_1+in_2$ for a resonator array in which the distance between the resonators is small compared to a wavelength.

array with $Q=20$ and $f=0.1$ and the result is shown in Fig. 17. The extreme values of the phase velocity are seen to be about $0.56c$ and $1.40c$ which are remarkably different from the free-field velocity. The attenuation at the peak is very large, corresponding to an attenuation of about 40 db per wavelength.

Measurements done on propagation of sound through one-dimensional arrays of resonators have also shown considerable attenuation at the resonance frequency.

A Single Resonator in an Infinite Wall

There are two cases of Helmholtz resonators in walls that permit a simple analysis. The first concerns a single resonator in a wall or even a group of resonators as long as the interaction between the individual resonators is negligible. This interaction can be neglected if the individual resonators in the group have resonance frequencies sufficiently far apart from each other. The other case corresponds to a group of resonators so close together as to render a practically uniform wall impedance. A typical example of such an arrangement is the perforated facing with partitioned air backing.

The analysis of a single resonator in a wall is very closely related to that of a resonator in a free field. The only differences are that independent of frequency the "blocked impedance" pressure at the aperture is now equal to $2P_0$, i.e., twice the incident pressure, and the radiation resistance of the aperture is twice that of the resonator in a free field. An analysis of the conditions for optimum design leads therefore to practically the same results as those already obtained in the previous section; only minor numerical differences occur. In the lists of optimum design values given above for the free-field resonator the figures for the wall resonator have also been presented within parentheses below the free-field values. These results follow from the original equations (20)–(22) if the C in the denominators is replaced by $2C$ and the C in the numerators is left unchanged. In the case when the frequency is kept constant and the resonator volume is varied, we therefore get the following relations between the optimum values for the wall and the free-field resonator:

$$(\sigma_{a1})_w = 2(\sigma_{a1})_f,$$

$$Q_{1w} = \frac{1}{2}Q_{1f},$$

$$T_{1w} = \frac{1}{\sqrt{2}}T_{1f}, \quad (47)$$

$$a_{1w} = \left(\frac{1}{2} \right)^{1/6} a_{1f},$$

where the subscripts w and f refer to the wall and free-field resonator, respectively.

The "design charts" in Figs. 11–16 can therefore be used also for the wall resonator if the following changes are considered:

1. Replace the maximum absorption cross section $\sigma_{a1} = (\lambda_0/4\pi)F$ by $\lambda_0^2/2\pi$.
2. Multiply T_1 and Q_1 by $1/\sqrt{2} = 0.707$.
3. Multiply a_1 by $(1/2)^{1/6} \approx 0.893$.
4. When applying the curves in Fig. 16, the optimum values a_1 , Q_1 , and T_1 have to be replaced by the corrected ones.

Example

A single resonator in a wall evidently contributes a relatively large absorption at resonance under optimum conditions. However, realizing that the corresponding Q value usually is of the order of 30 or 40, it is evident that the resonator under these conditions has a very limited use as an absorber. The Q value can be lowered either by introducing additional resistance in the hole or by using a non-optimum design. In the latter case the resonance absorption will be reduced more than the corresponding reduction in Q , as can be seen from Fig. 16. If the Q value is reduced by a factor of 2 by "mis-design," the resonance absorption cross section will be reduced approximately by a factor of 4. Despite this reduction in σ_a it may under certain conditions be advantageous to use a non-optimum design.

However, an optimum design can often be maintained with a lowered Q value if additional resistance is introduced into the resonator and the volume is increased. As a particular example let us consider a resonator with a resonance frequency 100 cps. We may assume $t/r_0 = 1$ and start with no additional resistance in the aperture, i.e., $\epsilon = 1$. The hole parameter h is then equal to 1.5, and from Fig. 13 we find an optimum equivalent sphere radius $a_1 = 15 \cdot 0.893 \approx 13.4$ cm, which corresponds to a maximum absorption cross section of $\lambda_0^2/2\pi$, i.e., approximately 18 sabines. The Q value is extremely high, however, $Q = 51$. If it is desired to reduce Q to, say, $Q = 10$, a hole parameter $h = 0.06$ and hence an additional hole resistance $\theta_{ap}c = 1.4$ rayls is required corresponding to $\epsilon = 36$ in Fig. 11. In order to maintain maximum resonance absorption, the resonator radius must be increased to $a_1 \approx 22.4$ cm, that is, the volume must be made about 4.7 times larger. If we decide not to change the volume, i.e., if we have $a/a_1 = 13.4/22.4 \approx 0.6$, the new Q value and absorption cross section are found from Fig. 16 to be $\sigma_a = 0.19 \times (\lambda_0^2/2\pi) \approx 3.5$ sabines and $Q_1 \approx 0.4 \cdot 10 = 4$. The aperture radius found from Fig. 12 is $r_0 \approx 0.26a \approx 2.7$ cm.

This example shows that undamped resonators under optimum conditions have such a high Q that they can, in general, be considered of little value as sound absorbers unless a proper damping is introduced.

Several Resonators in a Wall

When a resonator is made a member of a group of resonators in a wall its absorption cross section will be affected by the interaction from the other resonators. This effect is not easily evaluated except in the limiting case when the resonators are so close to each other that

the wall can be considered to have a uniform wall impedance. By comparison of this case with the theory based on no interaction, one can obtain some idea when the single resonator theory is applicable.

Let us thus consider n resonators arranged in a square array at a distance l apart. The resonance absorption cross section for a single resonator found from the above analysis is $\sigma_a = \lambda_0^2/2\pi = A_0/\theta_i$ (λ_0 = resonance wavelength, θ_i = internal specific resistance ratio of the resonator aperture, A_0 = area of aperture), and if the resonators are placed in the walls sufficiently far apart, they should act independently of each other yielding a total absorption of $n\sigma_a$ at resonance. If, on the other hand, the resonators in the square array are very close, the boundary can be considered to have a uniform impedance equal to $(l^2/A_0)\theta_i$ at resonance, and it is evident that the number of absorption units in this case will always be smaller than or at most equal to l^2 . As long as $l < \lambda_0/(2\pi)^{1/2}$, the total absorption evaluated in this way will therefore be smaller than when the resonators are considered independently. Correspondingly, when $l > \lambda/(2\pi)^{1/2}$ the uniform impedance assumption indicates a total (statistical) maximum resonance absorption larger than $n\sigma_a$; in fact, when $l^2 = A_0/\theta_i = \lambda_0^2/\pi$ it gives almost twice the value $n\sigma_a$. Although a thorough analysis of this problem is not carried through here, it seems reasonable to assume that $l = \lambda/(2\pi)^{1/2}$ represents approximately the limiting distance above which the resonators should be treated independently of each other. In other words, when the distance between the resonators exceeds approximately one-half wavelength, the total absorption should be determined as the sum of the individual absorption cross sections as calculated above. In the other limiting case when the resonators are very close to each other, it approaches the "perforated facing" type of a boundary and the design considerations here are not well applicable. Detailed studies of this latter configuration have been presented elsewhere.¹³

A Note on the Damping of Resonators

The additional aperture damping resistance $\theta_{ap}c$ referred to several times in the analysis can be obtained in practice in different ways. If the resistance is in the form of a screen or any other porous layer inserted in the aperture, then the total specific flow resistance of the porous layer should be equal to $\theta_{ap}c$. However, it is important to realize that if the screen is not inside or close up to the surface of the aperture but at a certain distance from it the effective resistance felt by the aperture is (considerably) smaller than the flow resistance of the porous layer. In this special case it is evident that the input resistance of the cavity varies from a maximum value to zero when the damping screen is moved from the position close up to the aperture to the back wall

¹³ J. Brillouin, Cahiers centre scient. et techn. batiment 31 (1949). Uno Ingard and R. H. Bolt, J. Acoust. Soc. Am. 23, 533 (1951).

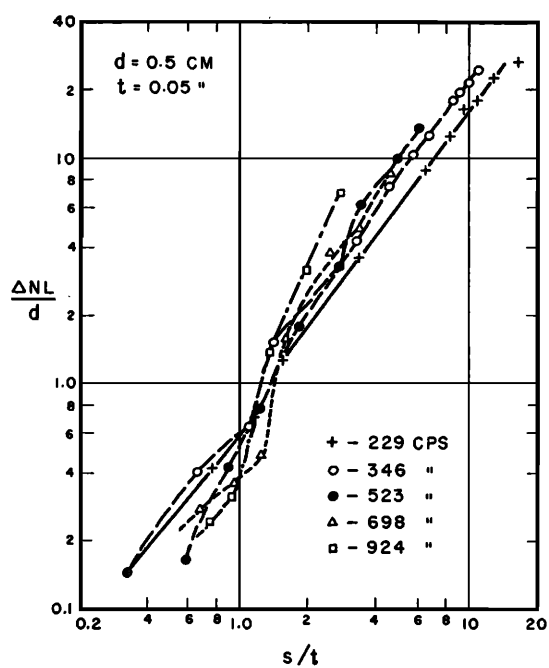


FIG. 18.

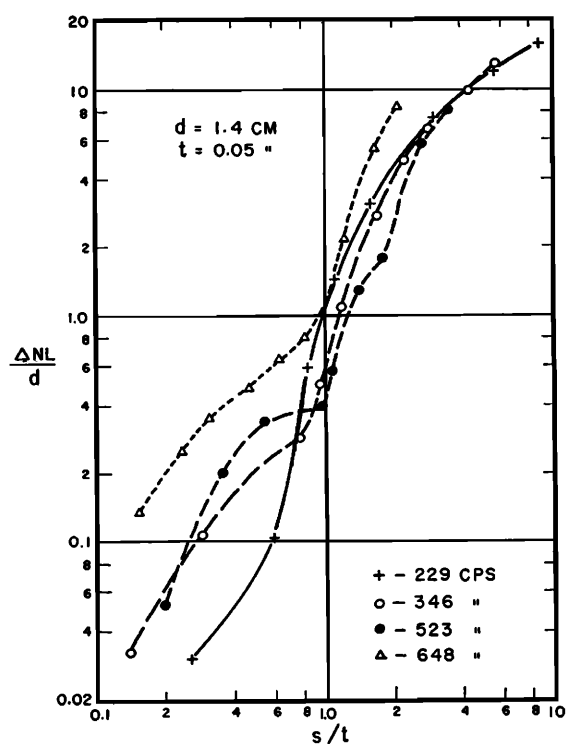


FIG. 20.

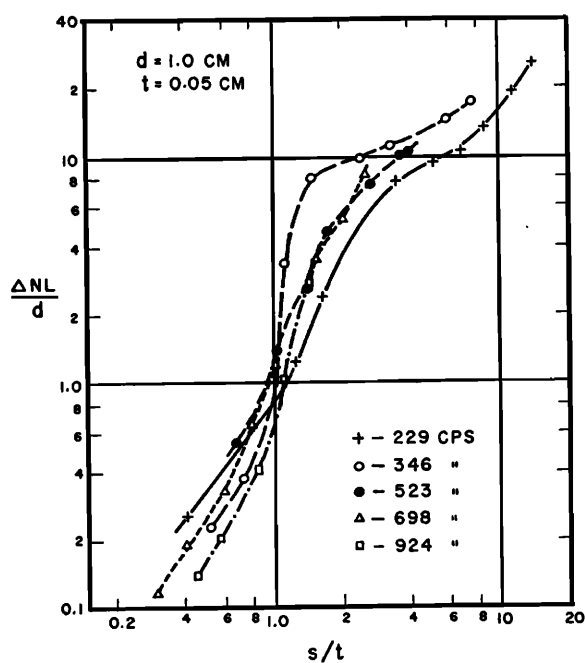


FIG. 19.

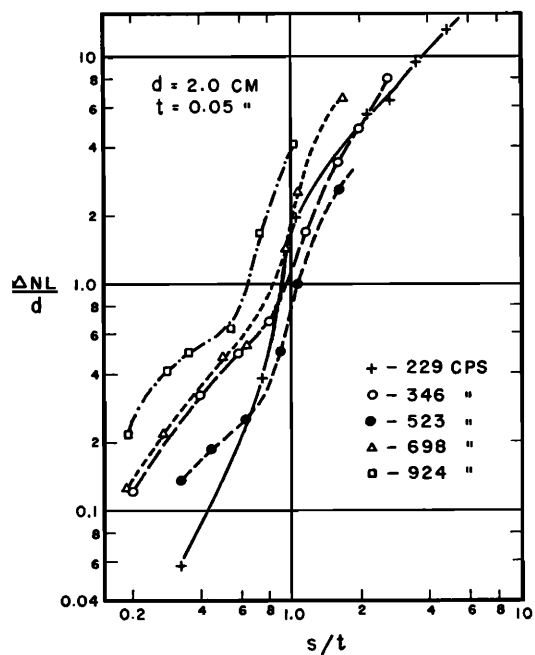


FIG. 21.

FIGS. 18-21. The nonlinear part of the resistance of an aperture, expressed in terms of a resistive end correction Δ_{nl} , and shown here as a function of the ratio s/t between the particle displacement amplitude in the aperture and the aperture thickness t .

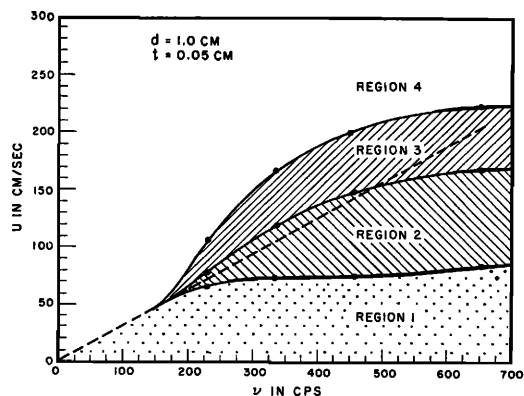


FIG. 22. Phase diagram of an aperture showing the regions of different types of circulations and turbulence. Regions 1 and 2 represent laminar acoustic streaming. Region 3 represents turbulent motion around the aperture, and in region 4 an acoustic jet is formed. The dashed line in the figure is $U = \omega t$, in which the particle displacement in the aperture equals the aperture thickness t .

of the cavity. The decrease is very rapid, and the slightest deviation of the screen can cause a considerable change in the resistance. A detailed study of this problem has been presented elsewhere.¹⁴

In the commonly used method of damping the resonator by filling the entire cavity with a porous material the specific aperture resistance will be $\theta_a \rho c \simeq (A_1/A_2)^{1/2} R t$ where R is the flow resistance of the material, t the depth of the cavity, and A_1/A_2 the ratio between the aperture area and the tube cavity area.

The Effect of the Number of Apertures on the Q Value

We have seen that the Q value for an undamped resonator with a single aperture in general is very high and that a decrease of Q often is desired if the resonator is to be used for absorption purposes. It should then be realized that in many cases a decrease of the Q value can be accomplished by splitting up the aperture into several small apertures without change of the resonance frequency.

Thus let the original aperture of area A be divided into n smaller apertures of area A_n . It is then clear from the expression for the resonance frequency (14) that, in order to get the same resonance frequency in the two cases, the area A_n should be

$$A_n = A/n, \text{ when } t \gg 0.96(A)^{1/2},$$

and

$$A_n = A/n^2, \text{ when } t \ll 0.96(A)^{1/2}.$$

In the first case, when the thickness t of the aperture is much larger than the mass end correction, the specific mass reactance of the aperture is approximately $k t \rho c$, and the resistance due to viscosity according to Eq. (16) approximately $2R_s t / r_0$. The ratio between the Q values

for the resonator with one aperture and that with n is then

$$Q/Q_n = n^{1/2}, \text{ when } t \gg 0.96(A)^{1/2}.$$

Correspondingly the specific mass reactance and the resistance of the aperture, when the thickness is small compared to the mass end correction, are approximately $k \delta \rho c$ and $4R_s$, respectively. The relation between the Q values for the resonators with one aperture and with n , Q , and Q_n , respectively, is then

$$Q_n = Q/n \quad t \ll 0.96(A)^{1/2}.$$

It has been tacitly assumed in this discussion that the apertures have been small enough so that the mass end correction $\delta \simeq 0.96(A)^{1/2}$ can be used. Furthermore, any interaction between the apertures has been disregarded.

VI. NONLINEAR EFFECTS

Studies of the nonlinear effects connected with sound propagation through an aperture have shown that both

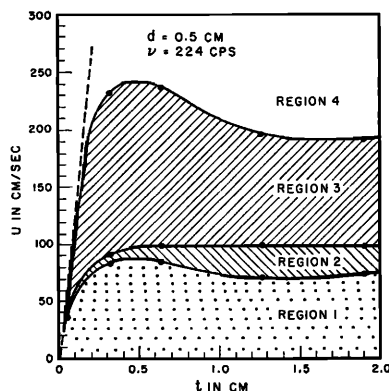


Fig. 23. Phase diagram for an aperture showing the regions of different types of acoustic streaming and turbulence in terms of the variables particle velocity U in the aperture and the aperture thickness t . The dashed line is $U = \omega t$ and the particle displacement in the aperture equals the thickness t .

the aperture resistance and the reactance depend on the intensity of the incident wave.¹⁵ In a previous paper the increase in resistance with intensity has been qualitatively and in certain cases quantitatively correlated with and explained by steady circulations and turbulence generated in the region of the aperture.¹⁶ Further measurements have since been made including studies also of the variation of the aperture reactance with sound intensity. The same measuring technique as described earlier was then used, and the acoustic impedance of circular apertures was determined for sound intensities from 60 to 140 db, and for aperture diameters ranging from 0.5 cm up to 2.0 cm.

Turbulence and Nonlinear Resistance

For apertures with a thickness t small enough to make the displacement amplitude s in the aperture of the same

¹⁴ Uno Ingard and R. H. Lyon, J. Acoust. Soc. Am. **25**, 854 (1953).

¹⁵ L. J. Sivian, J. Acoust. Soc. Am. **7**, 94 (1935).

¹⁶ Uno Ingard and S. Labate, J. Acoust. Soc. Am. **22**, 211 (1950).

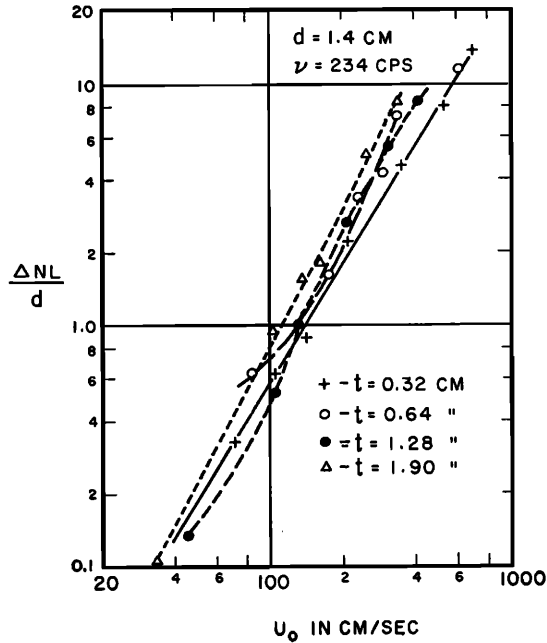


FIG. 24.

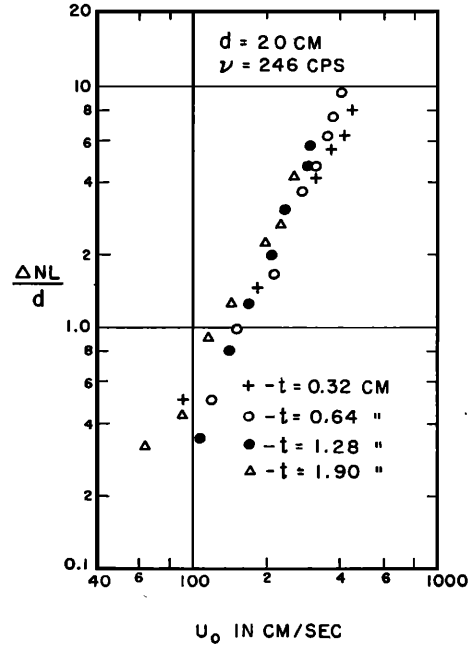


FIG. 26.

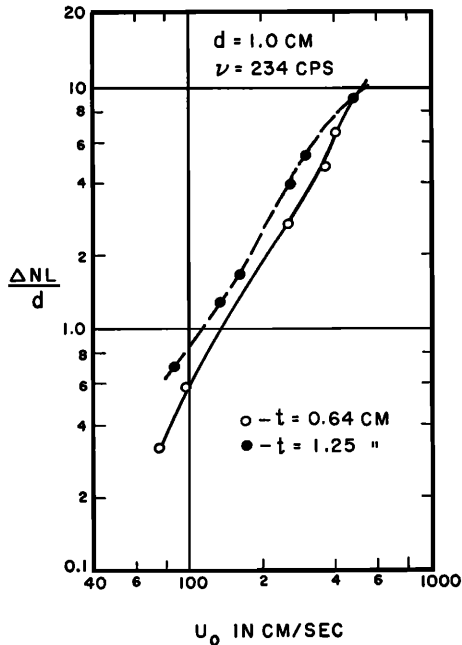


FIG. 25.

FIGS. 24-26. Nonlinear resistance as a function of particle velocity amplitude in the aperture.

order of magnitude as t , the nonlinear resistance seemed to be essentially determined by s/t . On the other hand, when $t \gg s$ the particle velocity amplitude seems to be the important variable.

In Figs. 18-21 examples of measured resistances for two thin apertures have been plotted as a function of s/t for different values of aperture diameter and frequency. The nonlinear part of the resistance has been

expressed in terms of a resistance end correction Δ_{nl} so that the total specific resistance of the aperture is expressed by the equation

$$\rho c \theta_{\text{total}} = \rho c (\theta_i + \theta_{nl}) = \frac{2R_s}{r_0} (t + \Delta_i + \Delta_{nl}). \quad (48)$$

The previous discussions indicated that the linear end correction for an aperture due to dissipation on the

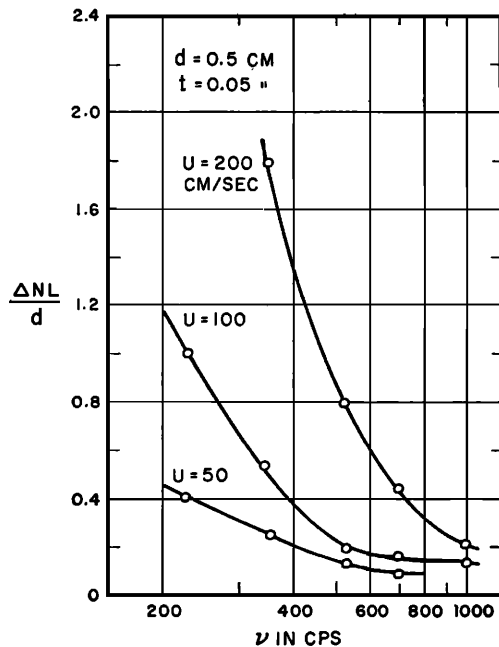


FIG. 27. This figure illustrates how the nonlinear resistance depends on the frequency when the particle velocity in the aperture is kept constant.

side walls of the plate should be set equal to $\Delta_t = 2r_0 = d$ and as a nondimensional form of the nonlinear part we have thus chosen $\Delta_{nl}/\Delta_t = \Delta_{nl}/d$. By this choice of variable it is seen that the curves in Figs. 18–21 are fairly well unified. It appears that when s/t is about equal to unity we have $\Delta_{nl} \approx \Delta_t$.

Several of the resistance curves indicate a marked change in slope in the region around $s/t = 1$. This can possibly be connected with onset of turbulence and jet effects in the aperture that appear to take place in this region. From the "phase diagrams" of an aperture, discussed in a previous paper,^{¶16} this onset is pictured as a function of frequency and particle velocity in the aperture as shown in Fig. 22. The thickness dependence is illustrated in Fig. 23. The different regions in these diagrams correspond to different states of acoustically generated dc streaming and jet effects described in some detail in reference 16. Regions 1 and 2 correspond to laminar streaming whereas region 4 represents jet formation. The intermediate region 3 is characterized by turbulent motion around the aperture, and the critical velocity corresponding to the lower boundary to this region is denoted in the following by U_c . Correspondingly we define a critical thickness $t_c = U_c/\omega$.

It is interesting to see in Fig. 23 that for sufficiently

¶ All particle velocities appearing in that paper are rms values. Furthermore, the following corrections should be made in that paper:

- Eq. (1): cancel factor of 2.
- Eq. (2): replace $\frac{1}{2}$ by $\frac{1}{4}$.
- Eq. (3): replace $\frac{1}{2}$ by $\frac{1}{4}$.
- Eq. (4): replace $\frac{1}{2}Fv$ [—] by Fv [—].

large wall thickness $t > t_c$, the turbulence onset occurs approximately at a constant value of U independent of aperture thickness t whereas for sufficiently small wall thicknesses $t < t_c$ the critical velocity becomes approximately proportional to the thickness. Figure 23 indicates that the constant of proportionality is ω since the straight line $U = \omega t$, shown dashed in the figure, seems to coincide with the turbulence boundary line.

The turbulence onset in a sound field is determined by critical values of the two variables involving particle velocity and frequency rather than by a single Reynolds number valid for steady flow. The experiments mentioned here indicate that for small t , $t < t_c$, the criterion for turbulence onset is approximately $s/t \approx 1$. In the experiments t_c was found to be of the order of 0.06 cm at least for frequencies below 500 cps.

When the thickness is increased, the velocity parameter seems to be the more important factor. Studies of the kind illustrated in Figs. 22 and 23 were made for four different values of diameters d at the frequency 234 cps. The critical velocity U_c for large t turned out to be practically independent of d and t ; the values observed ranged from 75 to 90 cm/sec. For values of $t > t_c$ we are thus tempted to consider the turbulence as essentially velocity determined.

The resistance measurements for thick apertures are also consistent with these observations on turbulence. For $t < t_c$ we have seen that the nonlinear resistance seems to be essentially dependent on s/t . On the other hand, for $t > t_c$, the velocity U turns out to be the more important variable as can be seen in Figs. 24–26. The nonlinear resistance plotted in this way as a function of velocity is almost independent of diameter and thickness of the aperture. Since the curves are practically straight lines we can set empirically

$$\Delta_{nl}/d \approx C(U/100)^n \quad (U \text{ in cm/sec}) \quad (t > t_c), \quad (49)$$

where C and n are constants that may depend on fre-

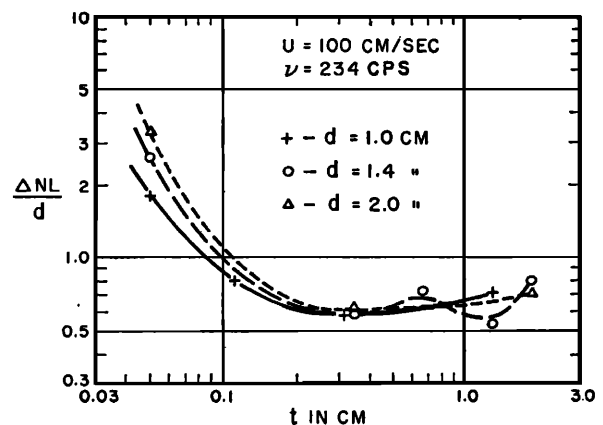


FIG. 28. The nonlinear resistance of an aperture as a function of the aperture thickness for constant value of the particle velocity U in the aperture.

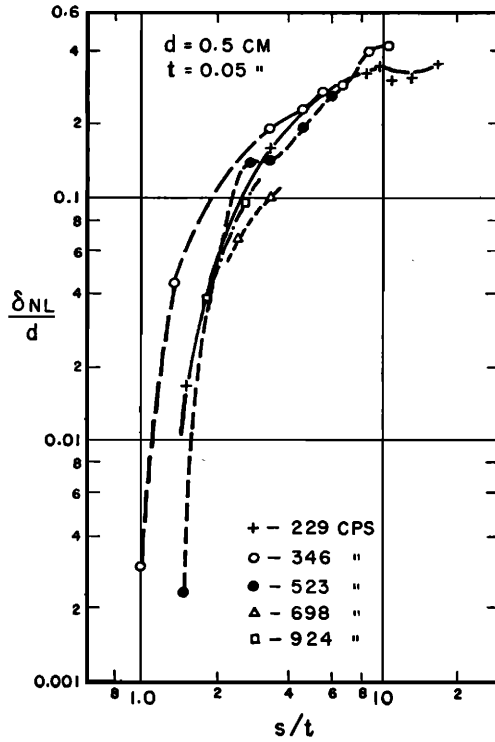


FIG. 29.

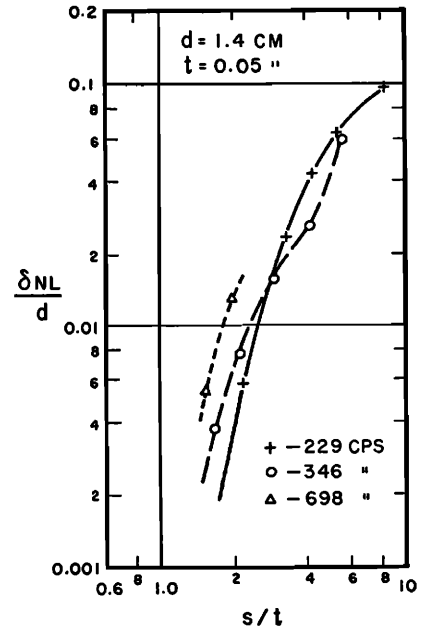


FIG. 31.

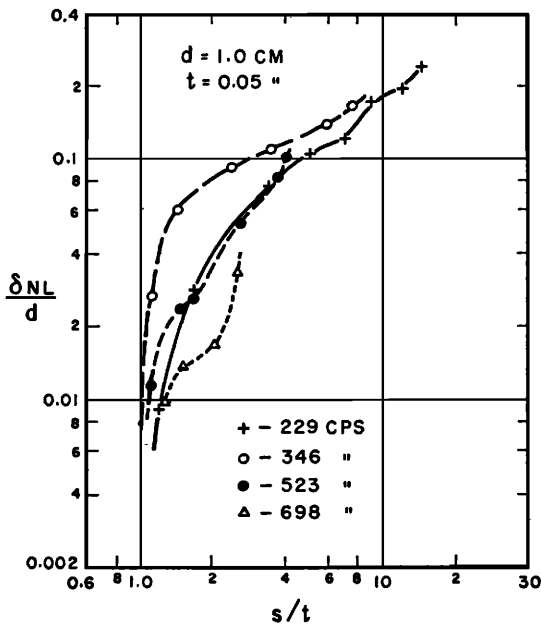


FIG. 30.

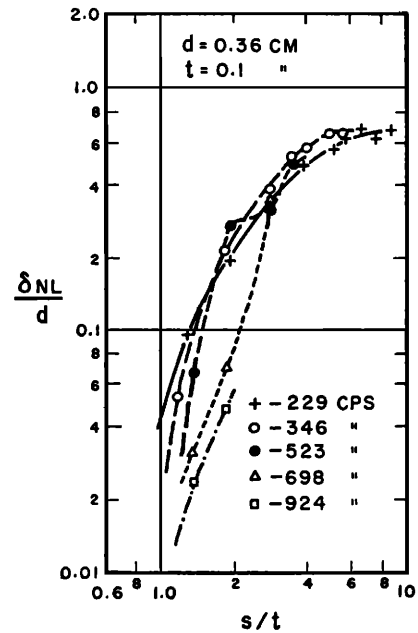


FIG. 32.

FIGS. 29-32. The decrease δ_{NL} of the mass end correction of an aperture as a function of s/t . (s =particle displacement in the aperture, t =aperture thickness.)

quency. The present measurements were done at one single frequency, 224 cps, at which we found $C=0.7$ and $n=1.7$.

In most measurements on the linear resistance it is quite difficult to avoid getting nonlinear contributions

into the results in particular at low frequencies and with thin apertures. This is illustrated in Figs. 27 and 28 where the nonlinear resistance for constant value of U has been plotted as a function of the frequency and thickness, respectively.

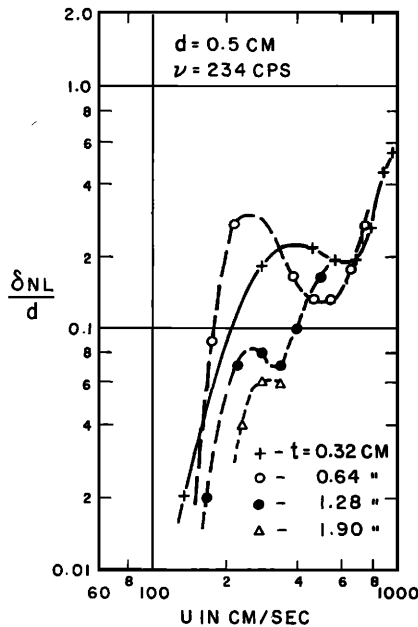


FIG. 33. The decrease δ_{nl} of the mass end correction of an aperture as a function of the velocity amplitude in the aperture.

The Nonlinear Reactance

The reactance of an aperture stays constant independent of the sound pressure level until turbulence is reached in the aperture region. In other words, the steady laminar circulations generated by the sound wave do not seem to affect the kinetic mass in the aperture field. The circulatory flow is apparently linearly superimposed upon the sound oscillations.

When turbulence is reached the mass reactance of the aperture starts to decrease. This nonlinear behavior is usually best observed with thin apertures for which the end correction is the major part of the mass. A number of curves showing the decrease in the kinetic mass, expressed as an equivalent end correction δ_{nl} , are shown in Figs. 29–32. The thickness of the aperture considered in these measurements is 0.05 cm. The curves have been plotted as a function of s/t , and it is seen that δ_{nl} is practically zero until $s/t \approx 1$ where turbulence sets in. The nonlinear reactance increases then rapidly and levels off when it approaches the linear end correction. For the aperture with $d = 0.36$ cm in Fig. 32, we see that δ_{nl}/d has reached a value of 0.7, i.e., very nearly the linear value for which $\delta_l/d \approx 0.8$.

Qualitatively the mechanism causing the decrease in the mass reactance can be thought of as a transfer of part of the kinetic mass (or kinetic energy) in the sound field around the aperture into turbulent motion and free boundary flow that breaks away from the aperture. The corresponding kinetic energy which ordinarily is reactive is thus lost and sooner or later dissipated by viscosity. In this sense there is a close tie between the nonlinear resistance and the reactance, the loss in mass reactance showing up as an increase in the resistance.

Westervelt¹⁷ has estimated the decrease in mass reactance for a small aperture to be $\frac{2}{3}$ of its linear value, a result that is of the right order of magnitude when compared with the maximum values in Figs. 29–32.

Measurements of the nonlinear mass reactance for thicker apertures with diameter 0.5 cm are shown in Fig. 33. The curves go through a maximum and a minimum before climbing up to this “saturation” value. This variation seems to be connected with the onset of a jet.

It is evident from the results thus presented that δ_{nl} decreases rather rapidly with increasing aperture diameter and thickness. The nonlinear reactance for the aperture with a diameter $d = 1.4$ cm is almost ten times smaller than the corresponding value for an aperture with diameter of 0.5 cm. In fact, measurements on an aperture with diameter of 2 cm gave no noticeable value of δ_{nl} in the frequency and intensity range considered in the other measurements.

It should be mentioned in this connection that a decrease of the mass reactance of an aperture is also obtained if the turbulence around the aperture is caused by external steady air flow past or through the aperture.¹⁸ The frequency of oscillation of resonators driven by a steady stream past or through the aperture is known to increase with the flow velocity within certain limits.¹⁹ This seems to be another aspect of the decrease of the mass reactance of the aperture due to turbulence.

Effect of Nonlinearity on the Design

From the results presented it is evident that the nonlinearity of the aperture impedance may affect the behavior of a resonator considerably, and the absorption derived from the linear design procedure may be in serious error at high sound intensities. In the analysis of this problem we consider only the effect of a nonlinear resistance. This is justified when the aperture thickness is much larger than the particle displacement in which case the decrease in aperture reactance with sound intensity is small.

The effect of nonlinearity can be incorporated in the design by letting the nonlinear resistance be absorbed in the parameter ϵ defined in Eq. (17). Considering the additional resistance θ_{nlp} in the expression for ϵ to be θ_{nlp} we get, using Eqs. (17) and (48),

$$\epsilon = 1 + \rho c \theta_{nl} / 4R_s = 1 + \frac{\Delta_{nl}}{d}. \quad (50)$$

In other words, the value of ϵ at intensities in the nonlinear range can be found directly from the results in the Figs. 24–26 or from the empirical formula (49). If

¹⁷ P. J. Westervelt, J. Acoust. Soc. Am. **23**, 347 (1951).

¹⁸ C. E. McAuliffe, S. M. thesis (Department of Naval Architecture and Marine Engineering, M.I.T., 1950).

¹⁹ A. B. C. Anderson, J. Acoust. Soc. Am. **24**, 675 (1952).

we thus use the experimentally found values $C=0.7$ and $n=1.7$, we can introduce $\Delta_{nl}/d=0.7(u/100)^{1.7}$ in Eq. (50).

For design purposes it is desirable to know Δ_{nl}/d in terms of the sound pressure level of the incident wave rather than the velocity in the aperture. The relation between the blocked pressure at the aperture and the velocity in the aperture at resonance is

$$p = \rho c (\theta_i + \theta_r) u, \quad (51)$$

where

$$\rho c \theta_i = 4R_s \left(1 + 0.5 \frac{t}{r_0} + 0.7 \left(\frac{u}{100} \right)^{1.7} \right).$$

The pressure $p = 2P_0$ for a resonator in a wall and P_0 for a resonator in a free field if P_0 = pressure of the incident wave. In the optimum design $\theta_i \approx \theta_r$, and the relation above becomes $p = 2\rho c \theta_i u$, or, using Eq. (51),

$$p = 8R_s u \left(1 + 0.5 \frac{t}{r_0} + 0.7 \left(\frac{u}{100} \right)^{1.7} \right). \quad (52)$$

From Eqs. (52), (49), and (50), the parameter ϵ can thus be calculated as a function of the sound pressure level of the incident waves. The results of the calculations are presented in Fig. 34.

To illustrate the effect of nonlinearity let us consider the example of the design of the 100-cps resonator discussed previously. It was found then that the radius for optimum design is $a=13.4$ cm and the Q value $Q \approx 51$. Now if the resonator is exposed to a wave of 140 db the ϵ value is not unity any more but $\epsilon \approx 66$. The hole parameter is then $h=0.035$, the corresponding optimum radius $a_1=28$ cm, and the Q value is $Q \approx 11$. If the original value of $a=13.4$ cm is used, we find $Q=0.22 \cdot 11=2.4$, i.e., the Q value is reduced from 51 to approximately 2.4 and the resonance absorption cross section from $\lambda_0^2/2\pi$ to $0.07\lambda_0^2/2\pi$. It is evident that the optimum "linear" design may be quite unsatisfactory and in the example shown here one should try to increase the sphere radius as close as possible to 28 cm to obtain a high-absorption cross section at 140 db.

It should be realized that the effect of nonlinearity analyzed above holds only at resonance and that the Q value obtained is essentially nothing else but a measure of the resistance of the aperture at resonance given by $Q = k_0(t+\delta)/\theta_i$. At frequencies sufficiently far from resonance, the effect of nonlinearity is usually minor since the reactance of the resonator prevents the velocity in the aperture from being very large.

In order to obtain the frequency dependence of the absorption cross section when the resonator is exposed to a sound wave of high intensity, one has to calculate the velocity $u = p/\zeta$ in the aperture for each frequency to be able to determine the nonlinear resistance. The calculation of the velocity is then perhaps most conveniently done by successive approximations. However, such calculations have shown that the frequency curve

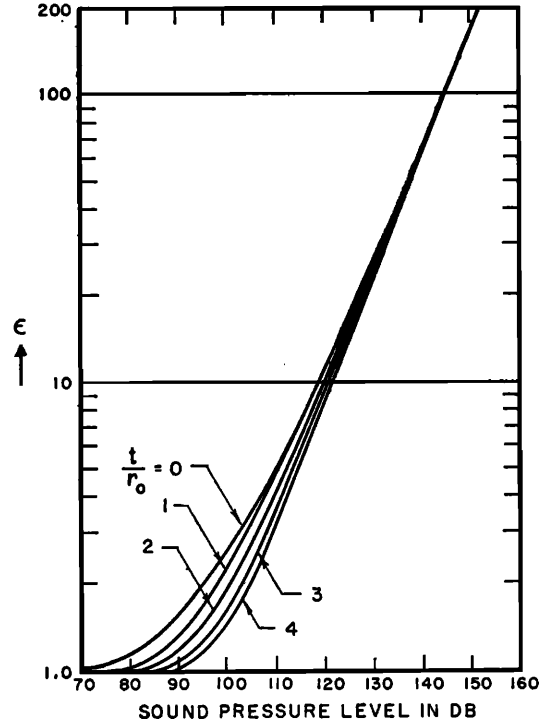


FIG. 34. The parameter ϵ as a function of the sound pressure level of the incident wave.

can very closely be approximated by the curve resulting from a combination of the absorption curves corresponding to the linear and the nonlinear Q value, respectively. In other words, to obtain the absorption curve at high intensities one uses the curve obtained from the linear treatment for frequencies on both sides sufficiently far from resonance and closes the curve over the frequency region in the neighborhood of resonance by the curve corresponding to Q_{nl} .

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APPENDIX I

The Mass End Correction of a Rectangular and a Circular Aperture in a Rectangular Tube

Rectangular Aperture

Replacing the aperture by an equivalent piston with uniform velocity amplitude U_0 , the expression for the pressure in the tubes is

$$p = \rho c \sum_m \sum_n a_{mn} e^{ik_{mn}x} \cos \pi \frac{my}{a} \cos \pi \frac{nz}{b}, \quad (i)$$

where

$$a_{mn} = \frac{v_{mn} U_0}{ab} \iint_{\text{piston}} \cos \left(\pi \frac{my}{a} \right) \cos \left(\pi \frac{nz}{b} \right) dy dz, \quad (ii)$$

$$v_{mn} = \begin{cases} \frac{1}{4} & \text{if } m=n \neq 0, \\ \frac{1}{2} & \text{if } m=0 \text{ } n \neq 0 \text{ or } m \neq 0 \text{ } n=0, \\ 1 & \text{if } m=0 \text{ } n=0, \end{cases}$$

a, b = sides of the rectangular tube.

For a rectangular concentric piston of dimension a_1 and b_1 , the integral in Eq. (ii) becomes

$$a_{mn} = 4\nu_{mn} \frac{\sin(\pi m a_1/a) \sin(\pi n b_1/b)}{mn\pi^2}.$$

Integrating the pressure in Eq. (i) over the area of the piston and dividing by the velocity, we find the mechanical impedance of the piston

$$z = 16a_1b_1\rho c \sum_{m=0} \sum_{n=0} \nu_{mn} \frac{g_{mn}}{\delta_{mn}},$$

where

$$g_{mn} = \left[\frac{\sin(\pi m a_1/a)}{\pi m a_1/a} \right]^2 \left[\frac{\sin(\pi n b_1/b)}{\pi n b_1/b} \right]^2 \frac{a_1 b_1}{ab}, \quad (\text{iii})$$

$$\delta_{mn} = (1 - \omega_{mn}^2/\omega^2)^{\frac{1}{2}},$$

$$\frac{\omega_{mn}}{2\pi} = \text{cut-off frequency of the } m, n \text{ mode,}$$

$$= c[(m/2a)^2 + (n/2b)^2]^{\frac{1}{2}}.$$

For frequencies well below the lowest normal mode we have $\delta_{mn} \simeq i\omega_{mn}/\omega$ and the mass reactance corresponding to the higher-order modes is $Z = (-i\omega)A_1\rho\delta_r$, where the end correction δ_r is

$$\delta_r = -\frac{2}{\pi}(ab)^{\frac{1}{2}} \sum'_m \sum'_n \nu_{mn} \frac{g_{mn}}{m^2 + n^2}. \quad (\text{iv})$$

The prime on the summation sign indicates that the term $m=0, n=0$ is excluded.

In the case of a square cross section the end correction δ_r has been calculated and plotted as a function of a_1/a and b_1/b in Fig. 4. The end correction in Eq. (iv) corresponds to the mass load on one side of the piston only, and the total end correction of an aperture is twice this value.

Circular Aperture

For a circular piston of radius r_0 , the integral (ii) becomes

$$a_{mn} = U_0 \frac{\nu_{mn}}{ab} \int_0^{2\pi} \int_0^{r_0} \cos\left(\pi m \frac{r_0}{a} \sin\varphi\right) \times \cos\left(\pi n \frac{r_0}{b} \cos\varphi\right) r dr d\varphi. \quad (\text{v})$$

If we split the product $\cos(-) \cos(-)$ into a sum of two $\cos$ terms and set

$$(m/a) \sin\varphi + (n/b) \cos\varphi = [(m/a)^2 + (n/b)^2]^{\frac{1}{2}} \sin(\varphi + \gamma),$$

where $\tan\gamma = na/mb$, the integration over φ in (v) leads to well-known integrals for the Bessel functions of zero order,

$$\begin{aligned} \int_0^{2\pi} \cos[r_1 \sin(\varphi + \gamma)] d\varphi &= \int_0^{2\pi} \cos[r_1 \sin(\varphi - \gamma)] d\varphi \\ &= \int_0^{2\pi} \cos(r_1 \sin\varphi) d\varphi = 2\pi J_0(r_1), \end{aligned}$$

in which

$$r_1 = \pi[(m/a)^2 + (n/b)^2]^{\frac{1}{2}} r_0.$$

Using $\int J_0(z)z dz = zJ_1(z)$ in the integration over r , we find

$$a_{mn} = 2U_0 \frac{\nu_{mn}}{ab} J_1(r_1).$$

Correspondingly, the mass end correction equivalent to the mass load on the circular piston in the (infinite) rectangular tube is

$$\delta_c = (\pi r_0^2)^{\frac{1}{2}} \frac{4}{\pi^{\frac{1}{2}}} \frac{(ab)^{\frac{1}{2}}}{r_0} \sum'_m \sum'_n \nu_{mn} \frac{J_1^2(r_1)}{\left[\frac{b}{a} m^2 + n^2 \frac{a}{b} \right]^{\frac{1}{2}}},$$

which for a square cross section $a=b$ reduces to

$$\delta_c = (\pi r_0^2)^{\frac{1}{2}} 4(\pi)^{\frac{1}{2}} \left(\frac{r_0}{a} \right)^2 \sum'_m \sum'_n \nu_{mn} \frac{J_1^2\left[\pi \frac{r_0}{a} (m^2 + n^2)^{\frac{1}{2}} \right]}{\left[\pi \frac{r_0}{a} (m^2 + n^2)^{\frac{1}{2}} \right]^3}.$$

APPENDIX II

The Interaction of Two Circular Apertures in a Cylindrical Tube

Let us start with the problem of finding the mass reactance of an eccentrically placed aperture as that shown in Table I. Assuming a uniform velocity distribution in the aperture, the problem is analogous to that of the radiation from an eccentrically located, plane circular piston into a circular tube. Using the coordinate system as shown in Fig. 5, the velocity potential in the tube is

$$\phi = \sum_m \sum_n A_{mn} e^{-\psi_{mn}} \sinh(\psi_{mn} + ik_{mn}x) \times J_m(b_{mn}r) \cos m\varphi, \quad (\text{i})$$

where

$$J_m'(b_{mn}R) = 0.$$

To find the constants A_{mn} the axial velocity in the plane of the piston is expanded in the series

$$(u_x)_{x=0} = \sum_m \sum_n a_{mn} J_m(b_{mn}r) e^{im\varphi}, \quad (\text{ii})$$

in which

$$a_{mn} = \frac{U_0}{2\pi \int_0^R J_m^2(b_{mn}r) r dr} \times \int_{\text{piston}} \int J_m(b_{mn}r) e^{-im\varphi} r dr d\varphi. \quad (\text{iii})$$

In order to perform the integration in (iii), the addition theorem²⁰ for the Bessel functions is used so that $J_m(b_{mn}r) \exp(-im\varphi)$ is expressed in terms of Bessel functions referred to an axis through the centrum of the orifice, corresponding to the new polar coordinates ρ and β ,

$$J_m(b_{mn}r) e^{-im\varphi} = \sum_{p=-\infty}^{+\infty} J_p(b_{mn}a) J_{m+p}(b_{mn}\rho) e^{-(p+m)\beta}. \quad (\text{iv})$$

Insertion of Eqs. (iv) in (iii) leads to

$$a_{mn} = 2U_0 \frac{r_0}{R} \frac{1}{b_{mn}R} \frac{J_{-m}(b_{mn}a) J_1(b_{mn}r_0)}{J_m^2(b_{mn}R) [1 - m^2/(b_{mn}R)^2]} \equiv U_0 g_{mn}. \quad (\text{v})$$

The coefficients A_{mn} are then given by

$$A_{mn} = \frac{a_{mn}}{-ik_{mn} \cosh \psi_{mn} e^{-\psi_{mn}}}.$$

From the total pressure P on the piston obtained by integration of the pressure $p = i\omega\phi$ over the piston we get the impedance

$$z = \frac{P}{U_0} = \pi r_0^2 \rho c \sum_m \sum_n F_{mn} \left(\frac{r_0}{R}, \frac{a}{R} \right) \frac{1}{\delta_{mn}} \tanh \psi_{mn}, \quad (\text{vi})$$

where

$$F_{mn} = \frac{4J_1^2(b_{mn}r_0) J_m^2(b_{mn}a)}{(b_{mn}R)^3 [1 - (m/b_{mn}R)^2] J_m^2(b_{mn}R)},$$

$$\delta_{mn} = [1 - (b_{mn}/k)^2]^{\frac{1}{2}} = i \frac{\omega_{mn}}{\omega} [1 - (\omega/\omega_{mn})^2]^{\frac{1}{2}},$$

ω_{mn} = cut-off frequency for (mn) mode.

The term $m=0, n=0$ represents the contribution from the plane wave which is purely resistive and of no interest now.

For frequencies much lower than the lowest cut-off frequency for the higher-order modes, i.e., $\omega \ll \omega_{mn}$, we have $\delta_{mn} \approx i\omega_{mn}/\omega \approx i\omega_{mn}/\omega$. Furthermore, considering an infinitely long tube so that $\tanh \psi_{mn} = 1$, the expression for the impedance in Eq. (vi) can be written

$$z = \pi r_0^2 (-i\omega) \rho \delta_{11},$$

²⁰ G. N. Watson, *Theory of Bessel Functions* (Cambridge University Press, Cambridge, 1944) p. 360.

where the mass end correction δ_{11} is

$$\delta_{11} = \frac{4}{(\pi)^{\frac{1}{2}}} (\pi r_0^2)^{\frac{1}{2}} \frac{1}{(r_0/R)} \times \sum'_m \sum'_n \frac{J_1^2(b_{mn}r_0)}{(b_{mn}R)^3 [1 - (m/b_{mn}R)^2]} \frac{J_m^2(b_{mn}a)}{J_m^2(b_{mn}R)}. \quad (\text{vii})$$

The prime on the summation sign means that the term $m=0, n=0$ is excluded. This expression reduces to the end correction for the concentric aperture when $a=0$. In that case m is always zero. The end correction δ_{11} has been calculated as a function of (r_0/R) for one value of the parameter $a/R=0.15$ and the result is shown in Fig. 6.

Consider now an additional aperture of the same size and at the same distance a from the center of the tube as shown in Fig. 5. If the pressure field of the first piston is p_1 and the second p_2 the total pressure is $p = p_1 + p_2$. If we denote the pressure caused by piston 1 at the surface of piston 2 by p_{12} the interaction impedance is

$$z_{12} = \frac{1}{U_2} \int_{S_2} p_{12} dS_2, \quad (\text{viii})$$

where U_2 is the velocity of the second piston.

The total sound field has the same form as that given in Eq. (i), but in determining the coefficients of the expansion the integration (iii) should be extended over both apertures. The pressure, using Eqs. (i) and (v), becomes

$$p_1 = \rho c U_1 \sum_m \frac{(g_{mn}) \sinh(\psi_{mn} + ik_{mn}x)}{\delta_{mn} \cosh \psi_{mn}} J_m(b_{mn}r) \cos m\varphi$$

and the same expression holds also for p_2 apart from a factor $(-1)^m$, i.e.,

$$p_2 = \rho c U_2 \sum_m (-1)^m \frac{g_{mn}}{\delta_{mn}}.$$

The interaction impedance thus becomes

$$z_{12} = (-i\omega) \pi r_0^2 \rho \delta_{12},$$

where, for $\omega \ll \omega_{mn}$,

$$\delta_{12} = \frac{4}{(\pi)^{\frac{1}{2}}} (\pi r_0^2)^{\frac{1}{2}} \sum'_m \sum'_n (-1)^m \times \frac{J_1^2(b_{mn}r_0) J_m^2(b_{mn}a)}{(b_{mn}R)^3 [1 - (m/b_{mn}R)^2]} \frac{1}{J_m^2(b_{mn}R)}.$$

The primes on the summation sign mean that the term $m=0, n=0$ is excluded. The δ_{12} has been plotted in Fig. 6 as a function of r_0/R to one value of $a/R=0.15$.